जब तक आपको यह परीक्षण पुस्तिका खोलने को न कहा जाए तब तक न खोलें

T.B.C.: FIAS-SFG26 **परीक्षण पुस्तिका अनुक्रम**

परीक्षण पुस्तिका

सामान्य अध्ययन

A

पूर्णांक : 100 **समय: एक घंटे**

अनुदेश

1. परीक्षा प्रारम्भ होने के तुरन्त बाद, आप इस परीक्षण पुस्तिका की पड़ताल अवश्य कर लें कि इसमें कोई बिना छपा, फटा या छूटा हुआ पृष्ठ अथवा प्रश्नांश, आदि न हो। यदि ऐसा है, तो इसे सही परीक्षण पुस्तिका से बदल लीजिए।
2. **कृपया ध्यान रखें कि OMR उत्तर-पत्रक में, उचित स्थान पर, रोल नम्बर और परिक्षण पुस्तिका अनुक्रम A, B, C या D को, ध्यान से एवं बिना किसी चूक या विसंगति के भरने और कूटबद्ध करने की जिम्मेदारी उम्मीदवार की है। किसी भी प्रकार की चूक/विसंगति की स्थिति में उत्तर -पत्रक निरस्त कर दिया जाएगा ।**
3. इस परीक्षण पुस्तिका पर साथ में दिए गए कोष्ठक में आपको अपना अनुक्रमांक लिखना है। परीक्षण पुस्तिका पर **और कुछ न** लिखें।

नाम:
ईमेल:
मोबाइल:

4. इस परीक्षण पुस्तिका में **50** प्रश्नांश (प्रश्न) हैं। प्रत्येक प्रश्नांश **अंग्रेज़ी और हिंदी** में छपा है। प्रत्येक प्रश्नांश में चार प्रत्युत्तर (उत्तर) दिए गए हैं। इनमें से एक प्रत्युतर को चुन लें, जिसे आप उत्तर-पत्रक पर अंकित करना चाहते हैं। यदि आपको ऐसा लगे कि एक से अधिक प्रत्युत्तर सही हैं, तो उस प्रत्युत्तर को अंकित करें जो आपको सर्वोत्तम लगे। किसी भी स्थिति में, प्रत्येक प्रश्नांश के लिए **केवल एक ही** प्रत्युत्तर चुनना है।
5. आपको अपने सभी प्रत्युत्तर अलग से दिए गए उत्तर-पत्रक पर ही अंकित करने हैं। उत्तर-पत्रक में दिए गए निर्देश देखिए।
6. ***सभी*** प्रश्नांशों *के अंक समान हैं।*
7. इससे पहले कि आप परीक्षण पुस्तिका के विभिन्न प्रश्नांशों के प्रत्युत्तर उत्तर-पत्रक पर अंकित करना शुरू करें, आपको प्रवेश प्रमाण-पत्र के साथ प्रेषित अनुदेशों के अनुसार कुछ विवरण उत्तर-पत्रक में देने हैं।
8. आप अपने सभी प्रत्युत्तरों को उत्तर-पत्रक में भरने के बाद तथा परीक्षा के समापन पर केवल उत्तर-पत्रक अधीक्षक को सौंप दें। आपको अपने साथ परीक्षण पुस्तिका ले जाने की अनुमति है।
9. कच्चे काम के लिए पत्रक परीक्षण पुस्तिका के अंत में संलग्न है।
10. **गलत उत्तरों के लिए दंड :**

वस्तुनिष्ठ प्रश्न-पत्रों में उम्मीदवार द्वारा दिए गए गलत उत्तरों के लिए दंड दिया जाएगा।

(i) प्रत्येक प्रश्न के लिए चार वैकल्पिक उत्तर हैं। उम्मीदवार द्वारा प्रत्येक प्रश्न के लिए दिए गए एक गलत उत्तर के लिए प्रश्न हेतु नियत किए गए अंकों का एक-तिहाई दंड के रूप में काटा जाएगा।

(ii) यदि कोई उम्मीदवार एक से अधिक उत्तर देता है, तो इसे गलत उत्तर माना जाएगा, यद्यपि दिए गए उत्तरों में से एक उत्तर सही होता है, फिर भी उस प्रश्न के लिए उपर्युक्तानुसार ही, उसी तरह का दंड दिया जाएगा।

(iii) यदि उम्मीदवार द्वारा कोई प्रश्न हल नहीं किया जाता है, अर्थात उम्मीदवार द्वारा उत्तर नहीं दिया जाता है, तो उस प्रश्न के लिए कोई दंड नहीं दिया जाएगा।

नोट: प्रश्न का उत्तर दें तथा अभ्यर्थी आत्मविश्वास स्तर (100% / 75% / 50% / 0%) का उपयोग करके उत्तर के प्रति अपनी निश्चितता का स्तर दर्ज कर सकते हैं। आप प्रत्येक प्रश्न के नीचे उपयुक्त आत्मविश्वास स्तर का चयन कर सकते हैं। इससे उन प्रश्नों की पुनरावृत्ति (रिवीजन) में सहायता मिलेगी, जिनमें उत्तर सही होने के बावजूद आप अनिश्चित थे।

जब तक आपको यह परीक्षण पुस्तिका खोलने को न कहा जाए तब तक न खोलें

Q.1) Consider the following statements:

Statement I: Studies indicate that carbon dioxide emissions from cement industry account for more than 5% of global carbon emissions.

Statement II: Silica-bearing clay is mixed with limestone while manufacturing cement.

Statement III: Limestone is converted into lime during clinker production for cement manufacturing.

Which one of the following is correct in respect of the above statements?

a) Both Statement II and Statement III are correct and both of them explain Statement I
b) Both Statement II and Statement III are correct but only one of them explains Statement I
c) Only one of the Statements II and III is correct and that explains Statement I
d) Neither Statement II nor Statement III is correct

100%		75%		50%		0%	

Q.2) Which of the following is/are anthropogenic sources of mercury pollution?

I. Artisanal and small-scale gold mining
II. Volcanic eruptions
III. Coal-fired power plants
IV. Burning of municipal and medical waste

Select the correct answer using the code given below.

a) I and III only
b) II only
c) I, III, and IV only
d) I, II, III and IV

100%		75%		50%		0%	

Q.3) With reference to microplastics, consider the following statements:

I. These are defined as shreds of plastic less than 5 millimeters in length
II. All microplastics presently come from the slow disintegration of larger plastic products.
III. Recent studies, in humans, showed that microplastics can cross placental barriers, which would disrupt the normal fetal growth and development.
IV. As per the recent WHO guidelines, in drinking water, microplastics up to 10,000 particles/L are considered as safe for drinking.

Which of the statements given above are correct?

a) I and III only
b) II, III and IV
c) I, II and III
d) I, II and IV

100%		75%		50%		0%	

Q.4) 'Carbosulfan' and 'Fenthion', which were seen recently in the news, are:

a) Chemical pesticides/insecticides added to Annex III of the Rotterdam Convention.
b) Prohibited chemical preservatives in processed foods under the Food Safety and Standards Authority of India regulations.
c) Primary chemical markers to identify the geographical origin of Basmati rice.
d) Bio-fertilizers approved in India for promotion of organic farming.

100%		75%		50%		0%	

Q.5) In the context of thermal pollution, as the temperature of a water body increases, the pH value of the water will generally:

a) Increase
b) Decrease
c) Remain unchanged
d) Increases initially and then decreases

100%		75%		50%		0%	

Q.1) निम्नलिखित कथनों पर विचार कीजिए:

कथन-I: अध्ययनों से सूचित होता है कि सीमेंट उद्योग से उत्सर्जित कार्बन डाइऑक्साइड का वैश्विक कार्बन उत्सर्जन में उसके 5% से अधिक का योगदान है।

कथन II: सीमेंट का निर्माण करते समय चूना-पत्थर में सिलिका-युक्त मृत्तिका मिलाई जाती है।

कथन III: सीमेंट निर्माण हेतु क्लिंकर उत्पादन के दौरान चूना-पत्थर, चूना में बदल जाता है।

उपर्युक्त कथनों के संबंध में निम्नलिखित में से कौन सा कथन सही है?

a) कथन II और कथन III दोनों सही हैं और दोनों ही कथन I की व्याख्या करते हैं।
b) कथन II और कथन III दोनों सही हैं, लेकिन उनमें से केवल एक ही कथन I की व्याख्या करता है।
c) कथन II और III में से केवल एक ही सही है और वह कथन I की व्याख्या करता है।
d) न तो कथन II और न ही कथन III सही है।

100%		75%		50%		0%	

Q.2) निम्नलिखित में से कौन-से पारे के प्रदूषण के मानवजनित स्रोत हैं?

I. छोटे पैमाने पर सोने का खनन
II. ज्वालामुखी विस्फोट
III. कोयले से चलने वाले विद्युत संयंत्र
IV. नगरपालिका और चिकित्सा अपशिष्ट का दहन

नीचे दिए गए कूट का उपयोग करके सही उत्तर चुनिए।

a) केवल I और III
b) केवल II
c) केवल I, III और IV
d) I, II, III और IV

100%		75%		50%		0%	

Q.3) माइक्रोप्लास्टिक के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. इन्हें 5 मिलीमीटर से कम लंबाई के प्लास्टिक के कणों के रूप में परिभाषित किया जाता है।
II. वर्तमान में सभी माइक्रोप्लास्टिक बड़े प्लास्टिक उत्पादों के मंद विघटन से उत्पन्न होते हैं।
III. मनुष्यों पर किए गए हाल के अध्ययनों से पता चला है कि माइक्रोप्लास्टिक गर्भनाल अवरोधों (प्लेसेंटल बैरियर) को पार कर सकते हैं, जिससे भ्रूण के सामान्य विकास और वृद्धि में बाधा उत्पन्न हो सकती है।
IV. विश्व स्वास्थ्य संगठन (WHO) के हालिया दिशानिर्देशों के अनुसार, पेयजल में 10,000 कण/लीटर तक के माइक्रोप्लास्टिक को पीने के लिए सुरक्षित माना जाता है।

उपर्युक्त कथनों में से कौन से सही हैं?

a) केवल I और III
b) II, III और IV
c) I, II और III
d) I, II और IV

100%		75%		50%		0%	

Q.4) हाल ही में चर्चित 'कार्बोसल्फान' और 'फेंथियन' हैं:

a) रॉटरडैम कन्वेंशन के परिशिष्ट III में शामिल रासायनिक कीटनाशक।
b) भारतीय खाद्य सुरक्षा एवं मानक प्राधिकरण के नियमों के तहत प्रसंस्कृत खाद्य पदार्थों में निषिद्ध रासायनिक परिरक्षक।
c) बासमती चावल के भौगोलिक मूल की पहचान करने वाले प्राथमिक रासायनिक संकेतक।
d) जैविक खेती को बढ़ावा देने के लिए भारत में स्वीकृत जैव उर्वरक।

100%		75%		50%		0%	

Q.5) ऊष्मीय प्रदूषण के संदर्भ में, किसी जल निकाय का तापमान बढ़ने पर, जल का pH मान सामान्यतः:

a) बढ़ेगा
b) घटेगा
c) अपरिवर्तित रहेगा
d) पहले बढ़ेगा और फिर घटेगा

100%		75%		50%		0%	

Q.6) Consider the following statements:

Statement I: Phthalates are mainly used as plasticizers to increase the durability of plastics.

Statement II: Phthalates form permanent chemical bonds with polymer chains of plastic.

Which one of the following is correct in respect of the above statements?

a) Both Statement I and Statement II are correct and Statement II explains Statement I.
b) Both Statement I and Statement II are correct but Statement II does not explain Statement I.
c) Statement I is correct but Statement II is not correct.
d) Statement I is not correct but Statement II is correct.

100%		75%		50%		0%	

Q.7) With reference to the World Air Quality Report 2024–25, consider the following statements:

I. Delhi has been ranked as the most polluted national capital in the world in terms of air quality.
II. Byrnihat city in Meghalaya is the cleanest city in India in terms of air quality.
III. The report uses PM2.5 as the primary metric for air quality reporting.

Which of the statements given above are correct?

a) I and II only
b) II and III only
c) I and III only
d) I, II and III

100%		75%		50%		0%	

Q.8) In the context of India, consider the following statements:

Statement I: As per National Ambient Air Quality Standards, unlike some other pollutants, tropospheric ozone levels are monitored in an eight-hour or one-hour average period, instead of annual or 24-hour averages.

Statement II: Tropospheric Ozone is highly reactive.

Which one of the following is correct in respect of the above statements?

a) Both Statement I and Statement II are correct and Statement II explains Statement I.
b) Both Statement I and Statement II are correct but Statement II does not explain Statement I.
c) Statement I is correct but Statement II is incorrect.
d) Statement I is incorrect but Statement II is correct.

100%		75%		50%		0%	

Q.9) The National Level Pollution Response Exercise (NATPOLREX) is conducted by:

a) Central Pollution Control Board.
b) National Environmental Engineering Research Institute (NEERI).
c) Indian Coast Guard.
d) Ministry of Environment, Forest and Climate Change.

100%		75%		50%		0%	

Q.10) With reference to the Wassenaar Arrangement, consider the following statements:

I. It is a multilateral export control regime headquartered in Hague, Netherlands.
II. It promotes transparency and greater responsibility in the transfer of conventional arms and dual-use goods and technologies between member countries.
III. India is a member of the Arrangement.

Which of the statements given above is/are correct?

a) I and II
b) II and III
c) I and III
d) II only

100%		75%		50%		0%	

Q.6) निम्नलिखित कथनों पर विचार कीजिए:

कथन I: प्लास्टिक की मजबूती बढ़ाने के लिए थैलेट का मुख्य रूप से प्लास्टिसाइज़र के रूप में उपयोग किया जाता है।

कथन II: थैलेट प्लास्टिक की बहुलक श्रृंखलाओं के साथ स्थायी रासायनिक बंध बनाते हैं।

उपर्युक्त कथनों के संबंध में निम्नलिखित में से कौन सा कथन सही है?

a) कथन I और कथन II दोनों सही हैं और कथन II, कथन I की व्याख्या करता है।

b) कथन I और कथन II दोनों सही हैं, लेकिन कथन II, कथन I की व्याख्या नहीं करता है।

c) कथन I सही है, लेकिन कथन II सही नहीं है।

d) कथन I सही नहीं है, लेकिन कथन II सही है।

100%		75%		50%		0%	

Q.7) विश्व वायु गुणवत्ता रिपोर्ट 2024-25 के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. वायु गुणवत्ता के मामले में दिल्ली को विश्व की सबसे प्रदूषित राष्ट्रीय राजधानी का दर्जा दिया गया है।

II. मेघालय का बर्नीहाट शहर वायु गुणवत्ता के मामले में भारत का सबसे स्वच्छ शहर है।

III. रिपोर्ट में वायु गुणवत्ता रिपोर्टिंग के लिए PM2.5 को प्राथमिक मापदंड के रूप में उपयोग किया गया है।

उपर्युक्त कथनों में से कौन से सही हैं?

a) केवल I और II

b) केवल II और III

c) केवल I और III

d) I, II और III

100%		75%		50%		0%	

Q.8) भारत के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

कथन I: राष्ट्रीय परिवेशी वायु गुणवत्ता मानकों के अनुसार, कुछ अन्य प्रदूषकों के विपरीत, क्षोभ मंडलीय ओजोन स्तरों की निगरानी वार्षिक या 24 घंटे के औसत के बजाय आठ घंटे या एक घंटे की औसत अवधि में की जाती है।

कथन II: क्षोभमंडलीय ओजोन अत्यधिक प्रतिक्रियाशील होती है।

उपर्युक्त कथनों के संबंध में निम्नलिखित में से कौन सा कथन सही है?

a) कथन I और कथन II दोनों सही हैं और कथन II, कथन I की व्याख्या करता है।

b) कथन I और कथन II दोनों सही हैं, लेकिन कथन II, कथन I की व्याख्या नहीं करता है।

c) कथन I सही है, लेकिन कथन II गलत है।

d) कथन I गलत है, लेकिन कथन II सही है।

100%		75%		50%		0%	

Q.9) राष्ट्रीय स्तर प्रदूषण प्रतिक्रिया अभ्यास (NATPOLREX) किसके द्वारा आयोजित किया जाता है?

a) केंद्रीय प्रदूषण नियंत्रण बोर्ड।

b) राष्ट्रीय पर्यावरण अभियांत्रिकी अनुसंधान संस्थान (NEERI)।

c) भारतीय तटरक्षक बल।

d) पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय।

100%		75%		50%		0%	

Q.10) वासेनार समझौते के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. यह एक बहुपक्षीय निर्यात नियंत्रण व्यवस्था है जिसका मुख्यालय हेग, नीदरलैंड में है।

II. यह सदस्य देशों के बीच पारंपरिक हथियारों और दोहरे उपयोग वाली वस्तुओं और प्रौद्योगिकियों के हस्तांतरण में पारदर्शिता और अधिक उत्तरदायित्व को बढ़ावा देता है।

III. भारत इस समझौते का सदस्य है।

उपर्युक्त कथनों में से कौन सा/से सही है/हैं?

a) I और II

b) II और III

c) I और III

d) केवल II

100%		75%		50%		0%	

Q.11) Consider the following statements:
1. Agricultural soils release nitrogen oxides into the environment.
2. Cattle release ammonia into the environment.
3. Poultry industry releases reactive nitrogen compounds into environment.

Which of the statements given above is/are correct?
a) 1 and 3 only
b) 2 and 3 only
c) 2 only
d) 1, 2 and 3

100%		75%		50%		0%	

Q.12) With reference to biomining and biocapping, consider the following statements:
I. Biomining involves using microorganisms to degrade waste and recover useful materials.
II. Biocapping involves covering waste dumps with layers of soil to prevent the release of pollutants.

Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.13) The Global Outlook Council on Water Investments is an initiative associated with:
a) International Water Management Institute
b) World Bank
c) G20
d) Wetland International

100%		75%		50%		0%	

Q.14) With reference to the Environment Protection (Management of Contaminated Sites) Rules, 2025, consider the following statements:
I. These rules have been notified under the Air (Prevention and Control of Pollution) Act, 1981
II. The rules provide a framework to identify and remediate chemically contaminated sites.
III. These rules exclude contamination of sites arising from radioactive waste.

Which of the statements given above are correct?
a) I and II only
b) II and III only
c) I, II and III
d) I and III only

100%		75%		50%		0%	

Q.15) With reference to the Armed Forces (Special Powers) Act, 1958, which of the following authorities has/have the power to declare a region as "disturbed area"?
I. Supreme Court of India
II. Central Government
III. Governor of the concerned State
IV. High Court of the concerned State

Select the correct answer using the code given below:
a) III only
b) III and IV
c) II and III
d) I and II

100%		75%		50%		0%	

Q.16) Dissolved Oxygen (DO) in water bodies is influenced by various physical and environmental factors. In this context, consider the following statements:
I. Under similar conditions, cold water can hold more dissolved oxygen than warm water.
II. Under similar temperature and pressure conditions, freshwater can hold more dissolved oxygen than seawater.
III. Compared to lower latitudes, dissolved oxygen levels are generally higher at high altitudes due to lower temperatures.
IV. Stagnant water bodies tend to have lower dissolved oxygen compared to flowing or turbulent water bodies.

Which of the statements given above are correct?
a) I, II and III only
b) I, II and IV only
c) III and IV only
d) I, II, III and IV

100%		75%		50%		0%	

Q.11) निम्नलिखित कथनों पर विचार कीजिए:
1. कृषि मृदाएँ पर्यावरण में नाइट्रोजन ऑक्साइड्स का उत्सर्जन करती हैं।
2. पशु पर्यावरण में अमोनिया का उत्सर्जन करते हैं।
3. कुक्कुट उद्योग पर्यावरण में प्रतिक्रियाशील नाइट्रोजन यौगिक का उत्सर्जन करते हैं।

उपर्युक्त कथनों में से कौन सा/से सही है/हैं?
a) केवल 1 और 3
b) केवल 2 और 3
c) केवल 2
d) 1, 2 और 3

100%		75%		50%		0%	

Q.12) बायोमाइनिंग और बायोकैपिंग के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:
I. बायोमाइनिंग में सूक्ष्मजीवों का उपयोग करके अपशिष्ट को विघटित करना और उपयोगी सामग्री को पुनः प्राप्त करना शामिल है।
II. बायोकैपिंग में प्रदूषकों के उत्सर्जन को रोकने के लिए अपशिष्ट स्थलों को मिट्टी की परतों से ढकना शामिल है।

उपर्युक्त कथनों में से कौन सा/से सही है/हैं?
a) केवल I
b) केवल II
c) I और II दोनों
d) न तो I और न ही II

100%		75%		50%		0%	

Q.13) जल निवेश पर वैश्विक दृष्टिकोण परिषद (ग्लोबल आउटलुक काउंसिल ऑन वाटर इन्वेस्टमेंट्स) किससे संबद्ध एक पहल है?
a) अंतर्राष्ट्रीय जल प्रबंधन संस्थान
b) विश्व बैंक
c) G20
d) वेटलैंड इंटरनेशनल

100%		75%		50%		0%	

Q.14) पर्यावरण संरक्षण (दूषित स्थलों का प्रबंधन) नियम, 2025 के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:
I. ये नियम वायु (प्रदूषण निवारण एवं नियंत्रण) अधिनियम, 1981 के अंतर्गत अधिसूचित किए गए हैं।
II. ये नियम रासायनिक रूप से दूषित स्थलों की पहचान और उपचार के लिए एक ढांचा प्रदान करते हैं।
III. ये नियम रेडियोधर्मी अपशिष्ट से उत्पन्न स्थलों के संदूषण को शामिल नहीं करते हैं।

उपर्युक्त कथनों में से कौन से सही हैं?
a) केवल I और II
b) केवल II और III
c) I, II और III
d) केवल I और III

100%		75%		50%		0%	

Q.15) सशस्त्र बल (विशेष शक्तियां) अधिनियम, 1958 के संदर्भ में, निम्नलिखित में से किस प्राधिकरण को किसी क्षेत्र को "अशांत क्षेत्र" घोषित करने का अधिकार है?
I. भारत का सर्वोच्च न्यायालय
II. केंद्र सरकार
III. संबंधित राज्य का राज्यपाल
IV. संबंधित राज्य का उच्च न्यायालय

नीचे दिए गए कूट का उपयोग करके सही उत्तर चुनिए:
a) केवल III
b) III और IV
c) II और III
d) I और II

100%		75%		50%		0%	

Q.16) जल निकायों में घुलित ऑक्सीजन (DO) विभिन्न भौतिक और पर्यावरणीय कारकों से प्रभावित होती है। इस संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:
I. समान परिस्थितियों में, ठंडा जल गर्म जल की तुलना में अधिक घुलित ऑक्सीजन धारण कर सकता है।
II. समान तापमान और दाब की स्थितियों में, ताजे जल में समुद्री जल की तुलना में अधिक घुलित ऑक्सीजन हो सकती है।
III. निम्न अक्षांशों की तुलना में, उच्च ऊंचाई पर तापमान कम होने के कारण घुलित ऑक्सीजन का स्तर आमतौर पर अधिक होता है।
IV. स्थिर जल निकायों में बहते या अशांत जल निकायों की तुलना में घुलित ऑक्सीजन का स्तर कम होता है।

उपर्युक्त कथनों में से कौन से सही हैं?
a) केवल I, II और III
b) केवल I, II और IV
c) केवल III और IV
d) I, II, III और IV

100%		75%		50%		0%	

Q.17) Recently, the government carried out reforms in the Disaster Management framework through the enactment of the Disaster Management (Amendment) Act, 2025. In this context, consider the following statements:

I. It empowers the state government to constitute Urban Disaster Management Authority (UDMA) in the state capital.

II. It shifts the responsibility of preparing national disaster management plans from the National Disaster Management Authority (NDMA) to the National Executive Committee.

III. It mandates the creation of a National Disaster Database incorporating risk assessments, mitigation strategies, and real-time disaster data.

Which of the statements given above are correct?

a) I, II and III
b) I and III only
c) II and III only
d) I and II only

100%		75%		50%		0%	

Q.18) With reference to pollution monitoring in India, consider the following information:

Index	**Implementing agency**	**Description**
I. National Air Quality Monitor-ing Programme (NAMP)	Central Pollution Control Board	It covers all the districts of India
II. National Air Quality Index (NAQI)	Commission for Air Quality Management in the National Capital Region and Adjoining Areas	It communic a-tes air quality in simple terms through One Number – One Color – One Descrip-tion.
III. System of Air Quality Forecasting and Research (SAFAR)	Indian Institute of Tropical Meteorology (IITM)	Apart from forecast-ing air quality it also provides forecasts on weather, emissions and UV radiation.

In which of the above rows is the given information correctly matched?

a) Only one
b) Only two
c) All the three
d) None

100%		75%		50%		0%	

Q.19) In the context of Environmental Impact Assessment (EIA) rules in India, consider the following projects/activities:

I. Starting a petroleum refining project

II. Construction of highway projects of strategic importance, which are within 100 km from the Line of Control.

III. Development of a coal-based thermal power plant with a capacity of 600 MW.

Which of the above projects are most likely to require prior environmental clearance?

a) I and III only
b) I and II only
c) II and III only
d) I, II and III

100%		75%		50%		0%	

Q.17) हाल ही में, सरकार ने आपदा प्रबंधन (संशोधन) अधिनियम, 2025 के अधिनियम के माध्यम से आपदा प्रबंधन ढांचे में सुधार किए हैं। इस संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. यह राज्य सरकार को राज्य की राजधानी में शहरी आपदा प्रबंधन प्राधिकरण (UDMA) गठित करने का अधिकार देता है।

II. यह राष्ट्रीय आपदा प्रबंधन योजनाओं को तैयार करने की जिम्मेदारी राष्ट्रीय आपदा प्रबंधन प्राधिकरण (NDMA) से राष्ट्रीय कार्यकारी समिति को स्थानांतरित करता है।

III. यह जोखिम आकलन, शमन रणनीतियों और वास्तविक समय के आपदा डेटा को शामिल करते हुए एक राष्ट्रीय आपदा डेटाबेस के निर्माण को अनिवार्य बनाता है।

उपर्युक्त कथनों में से कौन से सही हैं?

a) I, II और III

b) केवल I और III

c) केवल II और III

d) केवल I और II

100%		75%		50%		0%	

Q.18) भारत में प्रदूषण निगरानी के संदर्भ में निम्नलिखित जानकारी पर विचार कीजिए:

सूचकांक	**क्रियान्वयन एजेंसी**	**विवरण**
I. राष्ट्रीय वायु गुणवत्ता निगरानी कार्यक्रम (NAMP)	केंद्रीय प्रदूषण नियंत्रण बोर्ड	यह भारत के सभी जिलों को शामिल करता है
II. राष्ट्रीय वायु गुणवत्ता सूचकांक (NAQI)	राष्ट्रीय राजधानी क्षेत्र और आसपास के क्षेत्रों के लिए वायु गुणवत्ता प्रबंधन आयोग	यह वायु गुणवत्ता को सरल रूप में "एक संख्या – एक रंग – एक विवरण" के माध्यम से प्रस्तुत करता है
III. वायु गुणवत्ता पूर्वानुमान और अनुसंधान प्रणाली (SAFAR)	भारतीय उष्णकटिबंधीय मौसम विज्ञान संस्थान (IITM)	वायु गुणवत्ता के पूर्वानुमान के अतिरिक्त यह मौसम, उत्सर्जन तथा पराबैंगनी विकिरण का भी पूर्वानुमान प्रदान करता है

उपर्युक्त में से कितनी पंक्तियों में दी गई जानकारी सही सुमेलित है?

a) केवल एक

b) केवल दो

c) सभी तीन

d) कोई नहीं

100%		75%		50%		0%	

Q.19) भारत में पर्यावरण प्रभाव आकलन (EIA) नियमों के संदर्भ में, निम्नलिखित परियोजनाओं/गतिविधियों पर विचार करें:

I. पेट्रोलियम परिशोधन परियोजना की शुरुआत करना

II. सामरिक महत्व की राजमार्ग परियोजनाओं का निर्माण, जो नियंत्रण रेखा से 100 किमी के अंतर्गत हैं।

III. 600 मेगावाट की क्षमता वाले कोयला आधारित ताप विद्युत संयंत्र का विकास।

उपर्युक्त में से किन परियोजनाओं के लिए पूर्व पर्यावरणीय स्वीकृति की आवश्यकता होने की सर्वाधिक संभावना है?

a) केवल I और III

b) केवल I और II

c) केवल II और III

d) I, II और III

100%		75%		50%		0%	

Q.20) Which of the following best describes the objective of "Operation Black Forest", recently seen in the news?

a) An initiative aimed at reducing black carbon emissions by controlling large-scale forest fires.
b) A programme focused on improving livelihood opportunities for Denotified Tribes.
c) A coordinated drive to remove invasive plant species from Indian forest ecosystems.
d) A specialised anti-Naxal military operation conducted by Indian security forces.

100%		75%		50%		0%	

Q.21) Consider the following statements:

I. Carbon dioxide (CO_2) emissions in India are less than 0.5 t CO_2/capita.
II. In terms of CO_2 emissions from fuel combustion, India ranks second in the Asia-Pacific region.
III. Electricity and heat producers are the largest sources of CO_2 emissions in India.

Which of the statements given above is/are correct?

a) I and III only
b) II only
c) II and III only
d) I, II and III

100%		75%		50%		0%	

Q.22) Consider the following statements:

Plastic pollution has a negative health impact because the pollutants in plastic pollution can be:

I. Carcinogenic
II. Neurotoxicant
III. Endocrine disruptor

Which of the statements given above are correct?

a) I and II only
b) I and III only
c) II and III only
d) I, II and III

100%		75%		50%		0%	

Q.23) A contaminated industrial site contains complex pollutants that cannot be effectively degraded by the indigenous microbial population. To enhance the cleanup process, genetically modified bacteria are introduced into the soil to accelerate detoxification.

Which one of the following techniques is best described in the above scenario?

a) Biosparging
b) Bioaugmentation
c) Bioventing
d) Bioslurping

100%		75%		50%		0%	

Q.24) With reference to pollutants, their sources, and associated health impacts, consider the following information:

Pollutant	**Source**	**Associated Health Disorder**
I. Arsenic	Coal-based Thermal power plants	Knock-knee syndrome
II. Nitrates	Untreated sewage	Blue Baby Syndrome
III. Uranium	Phosphate fertilizers	Itai-Itai disease
IV. Fluoride	Aluminium smelting	Skeletal Fluorosis

In which of the above rows is the given information correctly matched?

a) Only one
b) Only two
c) Only three
d) All the four

100%		75%		50%		0%	

Q.20) हाल ही में चर्चित "ऑपरेशन ब्लैक फॉरेस्ट" का उद्देश्य निम्नलिखित में से कौन-सा सबसे उचित रूप से वर्णित करता है?

a) बड़े पैमाने पर वनाग्नि को नियंत्रित करके ब्लैक कार्बन उत्सर्जन को कम करने की पहल
b) अधिसूचित जनजातियों के लिए आजीविका के अवसरों में सुधार पर केंद्रित कार्यक्रम
c) भारतीय वनों से आक्रामक पादप प्रजातियों को हटाने हेतु समन्वित अभियान
d) भारतीय सुरक्षा बलों द्वारा संचालित एक विशेष नक्सल-विरोधी सैन्य अभियान

100%		75%		50%		0%	

Q.21) निम्नलिखित कथनों पर विचार कीजिए:

I. भारत में कार्बन डाइऑक्साइड (CO_2) उत्सर्जन 0.5 t CO_2 प्रति व्यक्ति से कम है।
II. ईंधन दहन से CO_2 उत्सर्जन के आधार पर, भारत एशिया-प्रशांत क्षेत्र में दूसरे स्थान पर है।
III. भारत में विद्युत और ताप उत्पादक, CO_2 उत्सर्जन के सबसे बड़े स्रोत हैं।

उपर्युक्त कथनों में से कौन-सा/से सही है/हैं?

a) केवल I और III
b) केवल II
c) केवल II और III
d) I, II और III

100%		75%		50%		0%	

Q.22) निम्नलिखित कथनों पर विचार कीजिए:

प्लास्टिक प्रदूषण का स्वास्थ्य पर नकारात्मक प्रभाव पड़ता है क्योंकि प्लास्टिक प्रदूषण में मौजूद प्रदूषक हो सकते हैं:

I. कैंसरकारी (कार्सिनोजेनिक)
II. तंत्रिका-विषाक्त (न्यूरो टॉक्सिकेंट)
III. अंतःस्रावी अवरोधक

उपर्युक्त कथनों में से कौन सा सही है?

a) केवल I और II
b) केवल I और III
c) केवल II और III
d) I, II और III

100%		75%		50%		0%	

Q.23) एक प्रदूषित औद्योगिक स्थल में जटिल प्रदूषक मौजूद होते हैं जिन्हें स्थानीय सूक्ष्मजीव समुदाय द्वारा प्रभावी रूप से अपघटित नहीं किया जा सकता। शोधन प्रक्रिया को तेज करने के लिए, विषहरण को तीव्र करने हेतु आनुवंशिक रूप से संशोधित जीवाणुओं को मृदा में प्रविष्ट कराया जाता है।

उपर्युक्त परिदृश्य में निम्नलिखित में से कौन-सी तकनीक का सर्वोत्तम वर्णन किया गया है?

a) बायोस्पार्जिंग
b) बायो ऑग्मेंटेशन
c) बायोवेंटिंग
d) बायोस्लर्पिंग

100%		75%		50%		0%	

Q.24) प्रदूषकों, उनके श्रोतों और संबंधित स्वास्थ्य प्रभावों के संदर्भ में, निम्नलिखित जानकारी पर विचार कीजिए:

प्रदूषक	**श्रोत**	**संबंधित स्वास्थ्य विकार**
I. आर्सेनिक	कोयला आधारित ताप विद्युत संयंत्र	नॉक-नी सिंड्रोम
II. नाइट्रेट्स	अनुपचारित सीवेज	ब्लू बेबी सिंड्रोम
III. यूरेनियम	फॉस्फेट उर्वरक	इटाई-इटाई रोग
IV. फ्लोराइड	एल्युमीनियम प्रगलन	स्केलेटल फ्लोरोसिस

उपर्युक्त में से कितनी पंक्तियों में दी गई जानकारी सही सुमेलित है?

a) केवल एक
b) केवल दो
c) केवल तीन
d) सभी चार

100%		75%		50%		0%	

Q.25) Which one of the following air pollutants combines with the haemoglobin of human blood and reduces its oxygen-carrying capacity, leading to suffocation and may cause even death?
a) Chlorofluorocarbon
b) Fly ash
c) Carbon monoxide
d) Sulphur dioxide

100%		75%		50%		0%	

Q.26) In the context of global environmental pollution, the term "Grasshopper Effect" refers to:
a) Rapid biodegradation of pollutants in tropical regions.
b) Accumulation of pollutants in soil due to agricultural practices.
c) Long-range transport of persistent pollutants through repeated evaporation and condensation cycles.
d) The sudden increase in local pollutant concentration due to the migratory patterns of insects.

100%		75%		50%		0%	

Q.27) With reference to fly ash, consider the following statements:
I. Its composition includes silicon dioxide (SiO_2), calcium oxide (CaO), and aluminum oxide (Al_2O_3).
II. ASHTRACK is a web portal and app for better management of fly ash produced by thermal power plants.
III. It can be utilized as a soil conditioner to improve the water-holding capacity of soils.
IV. Fly ash bricks manufacturing units in India are listed under the red category of Industries as per the Central Pollution Control Board classification of industries based on Pollution index.
Which of the statements given above are correct?
a) I and II only
b) III and IV only
c) I, II and III only
d) I, II, III and IV

100%		75%		50%		0%	

Q.28) With reference to Chromium, consider the following statements:
I. The leather tanning industry is a major source of chromium pollution.
II. Chromium contamination in agricultural soil reduces soil microbial activity.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.29) Under the E-Waste (Management) Amendment Rules, 2024, who among the following shall fix the highest and lowest price for exchange of extended producer responsibility (EPR) certificates on the platform for exchange or transfer of EPR certificates?
a) Bureau of Indian Standards (BIS)
b) Securities and Exchange Board of India (SEBI)
c) Central Pollution Control Board (CPCB)
d) The Energy and Resources Institute (TERI)

100%		75%		50%		0%	

Q.30) Consider the following statements regarding the World Food Programme (WFP):
I. Food and Agriculture Organization was instrumental in the establishment of the World Food Programme.
II It is one of the agencies of the United Nations.
III. Its operations are entirely funded through the voluntary contributions from donors.
IV. It was awarded the Nobel Peace Prize in 2020.
Which of the statements given above are correct?
a) I and III only
b) I, II and III only
c) II and IV only
d) I, II, III and IV

100%		75%		50%		0%	

Q.25) निम्नलिखित में से कौन सा वायु प्रदूषक मानव रक्त के हीमोग्लोबिन के संयोजित होकर ऑक्सीजन ले जाने की इसकी क्षमता को कम करता है, जिससे श्वासावरोध होता है और मृत्यु भी हो सकती है?

a) क्लोरोफ्लोरोकार्बन
b) फ्लाई ऐश
c) कार्बन मोनोऑक्साइड
d) सल्फर डाइऑक्साइड

100%		75%		50%		0%	

Q.26) वैश्विक पर्यावरण प्रदूषण के संदर्भ में, "ग्रासहॉपर इफेक्ट" शब्द का तात्पर्य है:

a) उष्णकटिबंधीय क्षेत्रों में प्रदूषकों का तेजी से जैव निम्नीकरण।
b) कृषि पद्धतियों के कारण मिट्टी में प्रदूषकों का संचय।
c) बार-बार वाष्पीकरण और संघनन चक्रों के माध्यम से लगातार प्रदूषकों का लंबी दूरी का परिवहन।
d) कीड़ों के प्रवासी पैटर्न के कारण स्थानीय प्रदूषकों की सांद्रता में अचानक वृद्धि।

100%		75%		50%		0%	

Q.27) फ्लाई ऐश के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. इसकी संरचना में सिलिकॉन डाइऑक्साइड (SiO_2), कैल्शियम ऑक्साइड (CaO), और एल्यूमीनियम ऑक्साइड (Al_2O_3) शामिल हैं।
II. एशट्रैक (ASHTRACK) थर्मल पावर प्लांट द्वारा उत्पादित फ्लाई ऐश के बेहतर प्रबंधन के लिए एक वेब पोर्टल और ऐप है।
III. इसका उपयोग मिट्टी की जलधारण क्षमता में सुधार के लिए मृदा संशोधक के रूप में किया जा सकता है।
IV. भारत में फ्लाई ऐश ईंटों की निर्माण इकाइयों को प्रदूषण सूचकांक के आधार पर उद्योगों के केंद्रीय प्रदूषण नियंत्रण बोर्ड वर्गीकरण के अनुसार उद्योगों की लाल श्रेणी के तहत सूचीबद्ध किया गया है।

उपर्युक्त कथनों में से कौन सा सही है?

a) केवल I और II
b) केवल III और IV
c) केवल I, II और III
d) I, II, III और IV

100%		75%		50%		0%	

Q.28) क्रोमियम के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. चमड़ा टैनिंग उद्योग क्रोमियम प्रदूषण का एक प्रमुख स्रोत है।
II. कृषि मृदा में क्रोमियम संदूषण मृदा सूक्ष्मजीव गतिविधि को कम कर देता है।

उपर्युक्त कथनों में से कौन-सा/से सही है/हैं?

a) केवल I
b) केवल II
c) I और II दोनों
d) न तो I और न ही II

100%		75%		50%		0%	

Q.29) ई-कचरा (प्रबंधन) संशोधन नियम, 2024 के अंतर्गत, निम्नलिखित में से कौन EPR (विस्तारित उत्पादक उत्तरदायित्व) प्रमाणपत्रों के विनिमय या हस्तांतरण के लिए प्लेटफार्म पर उनके उच्चतम और न्यूनतम मूल्य निर्धारित करेगा?

a) भारतीय मानक ब्यूरो (BIS)
b) भारतीय प्रतिभूति और विनिमय बोर्ड (SEBI)
c) केंद्रीय प्रदूषण नियंत्रण बोर्ड (CPCB)
d) ऊर्जा और संसाधन संस्थान (TERI)

100%		75%		50%		0%	

Q.30) विश्व खाद्य कार्यक्रम (WFP) के संबंध में निम्नलिखित कथनों पर विचार कीजिए:

I. खाद्य और कृषि संगठन ने विश्व खाद्य कार्यक्रम की स्थापना में महत्वपूर्ण भूमिका निभाई थी।
II यह संयुक्त राष्ट्र की एजेंसियों में से एक है।
III. इसके संचालन को पूरी तरह से दाताओं से स्वैच्छिक योगदान के माध्यम से वित्त पोषित किया जाता है।
IV. इसे 2020 में नोबेल शांति पुरस्कार से सम्मानित किया गया था।

उपर्युक्त कथनों में से कौन सा सही है?

a) केवल I और III
b) केवल I, II और III
c) केवल II और IV
d) I, II, III और IV

100%		75%		50%		0%	

Q.31) Which of the following is/are the possible consequence/s of heavy sand mining in river beds?
1. Decreased salinity in the river
2. Pollution of groundwater
3. Lowering of the water table
Select the correct answer using the code given below:
a) 1 only
b) 2 and 3 only
c) 1 and 3 only
d) 1, 2 and 3

100%		75%		50%		0%	

Q.32) Consider the following statements with respect to the C-FLOOD:
I. It is a web-based platform that provides two days' advance inundation forecasts up to the village level.
II. It provides information about storage levels of water in important reservoirs in the country.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.33) In the context of global risk management, which of the following best describes a 'Grey Rhino' event?
a) A highly improbable event that has massive impacts and is rationalized in hindsight.
b) A highly probable, high-impact threat which was ignored despite visible warning signs.
c) A gradual and insignificant risk that has minimal long-term impact.
d) A sudden, localized surge in pollution levels in an area due to industrial accidents.

100%		75%		50%		0%	

Q.34) Consider the following pollutants:
I. Sulphur dioxide (SO_2)
II. Nitrogen oxides (NO_x)
III. Methane (CH_4)
IV. Ammonia (NH_3)
Which of the pollutants given above are included under the Gothenburg Protocol which aims at reducing long-range transboundary air pollution?
a) I and II only
b) III and IV only
c) I, II and IV only
d) I, II, III and IV

100%		75%		50%		0%	

Q.35) With reference to Chemical Oxygen Demand (COD), consider the following statements:
I. It is a measure of the oxygen equivalent required to chemically oxidize total organic matter present in water.
II. COD values are generally lower than Biological Oxygen Demand (BOD) values for the same water sample.
III. In distilled water, the COD value is zero or nearly zero.
Which of the statements given above is/are correct?
a) I and III only
b) I only
c) II and III only
d) III only

100%		75%		50%		0%	

Q.36) The 'Global Drought Outlook, 2025' report was released by which of the following organizations?
a) United Nations Convention to Combat Desertification (UNCCD)
b) World Meteorological Organization (WMO)
c) Organisation for Economic Cooperation and Development (OECD)
d) Food and Agriculture Organization (FAO)

100%		75%		50%		0%	

Q.31) निम्नलिखित में से कौन-सा/से नदी तल में बहुत अधिक बालू खनन का/के संभावित परिणाम हो सकता है/सकते हैं?

1. नदी की लवणता में कमी
2. भौम जल का प्रदूषण
3. भौम जलस्तर का नीचे चले जाना

नीचे दिए गए कूट का प्रयोग कर सही उत्तर चुनिए :

a) केवल 1
b) केवल 2 और 3
c) केवल 1 और 3
d) 1, 2 और 3

100%		75%		50%		0%	

Q.32) C-FLOOD के संबंध में निम्नलिखित कथनों पर विचार कीजिए:

I. यह एक वेब-आधारित प्लेटफॉर्म है जो ग्राम स्तर तक दो दिनों के अग्रिम बाढ़ पूर्वानुमान प्रदान करता है।
II. यह देश के महत्वपूर्ण जलाशयों में जल के भंडारण स्तर की जानकारी प्रदान करता है।

उपर्युक्त कथनों में से कौन-सा/से सही है/हैं?

a) केवल I
b) केवल II
c) I और II दोनों
d) न तो I और न ही II

100%		75%		50%		0%	

Q.33) वैश्विक जोखिम प्रबंधन के संदर्भ में, निम्नलिखित में से कौन सा 'ग्रे राइनो' (Grey Rhino) घटना का सबसे अच्छा वर्णन करता है?

a) एक अत्यंत असंभावित घटना जिसका प्रभाव अत्यधिक होता है और जिसे बाद में तर्कसंगत बनाया जाता है।
b) एक अत्यधिक संभावित, उच्च-प्रभाव वाला खतरा जिसे स्पष्ट चेतावनी संकेतों के बावजूद नजरअंदाज कर दिया गया था।
c) एक क्रमिक और महत्वहीन जोखिम जिसका दीर्घकालिक प्रभाव न्यूनतम होता है।
d) औद्योगिक दुर्घटनाओं के कारण किसी क्षेत्र में प्रदूषण स्तर में अचानक, स्थानीय वृद्धि।

100%		75%		50%		0%	

Q.34) निम्नलिखित प्रदूषकों पर विचार कीजिए:

I. सल्फर डाइऑक्साइड (SO_2)
II. नाइट्रोजन ऑक्साइड (NO_x)
III. मीथेन (CH_4)
IV. अमोनिया (NH_3)

उपर्युक्त प्रदूषकों में से कौन से गोथेनबर्ग प्रोटोकॉल के तहत शामिल हैं जिसका उद्देश्य लंबी दूरी के ट्रांस बाउंड्री वायु प्रदूषण को कम करना है?

a) केवल I और II
b) केवल III और IV
c) केवल I, II और IV
d) I, II, III और IV

100%		75%		50%		0%	

Q.35) रासायनिक ऑक्सीजन मांग (COD) के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. यह जल में मौजूद कुल कार्बनिक पदार्थों को रासायनिक रूप से ऑक्सीकृत करने के लिए आवश्यक ऑक्सीजन समकक्ष का एक माप है।
II. एक ही जल के नमूने के लिए COD मान आम तौर पर जैविक ऑक्सीजन मांग (BOD) मानों से कम होते हैं।
III. आसुत जल में, COD मान शून्य या लगभग शून्य होता है।

उपर्युक्त कथनों में से कौन-सा/से सही है/हैं?

a) केवल I और III
b) केवल I
c) केवल II और III
d) केवल III

100%		75%		50%		0%	

Q.36) वैश्विक सूखा पूर्वानुमान (ग्लोबल ड्रॉट आउटलुक), 2025 रिपोर्ट निम्नलिखित में से किस संगठन द्वारा जारी की गई थी?

a) संयुक्त राष्ट्र मरुस्थलीकरण रोकथाम अभिसमय (UNCCD)
b) विश्व मौसम विज्ञान संगठन (WMO)
c) आर्थिक सहयोग और विकास संगठन (OECD)
d) खाद्य और कृषि संगठन (FAO)

100%		75%		50%		0%	

Q.37) With reference to Methane, consider the following statements:

I. It is a short-lived climate pollutant.

II. It is the second-largest contributor to global warming after carbon dioxide.

III. The Global Methane Pledge targets a minimum 60% reduction in global methane emissions from 2020 levels by 2030.

Which of the statements given above are correct?

a) I and II only

b) II and III only

c) I and III only

d) I, II and III

100%		75%		50%		0%	

Q.38) Consider the following substances:

I. Chlorofluorocarbons (CFCs).

II. Hydrochlorofluorocarbons (HCFCs).

III. Hydrofluorocarbons (HFCs).

IV. Carbon tetrachloride.

How many of the above are major Ozone Depleting Substances (ODS)?

a) Only one

b) Only two

c) Only three

d) All the four

100%		75%		50%		0%	

Q.39) The term 'LICOMK++', which was recently in the news, is best described as which of the following?

a) A high-performance supercomputing cluster for weather forecasting in the Himalayan region.

b) A high-resolution global ocean simulation system used for studying ocean dynamics.

c) An AI-based platform for predicting monsoon variability in tropical regions.

d) A deep-sea exploration vehicle designed for mapping ocean floors.

100%		75%		50%		0%	

Q.40) Consider the following pairs:

Area of conflict mentioned in News	Country where it is located
I. Deir el-Balah	Sudan
II. Khartoum	Somalia
III. Donetsk	Ukraine

How many of the pairs given above are correctly matched?

a) Only one

b) Only two

c) All the three

d) None

100%		75%		50%		0%	

Q.41) With reference to 'Direct Air Capture', an emerging technology, which of the following statements is/are correct?

I. It can be used as a way of carbon sequestration.

II. It can be a valuable approach for plastic production and in food processing.

III. In aviation, it can be a source of carbon for combining with hydrogen to create synthetic low-carbon fuel.

Select the correct answer using the code given below.

a) I and II only

b) III only

c) I, II and III

d) None of the above statements is correct

100%		75%		50%		0%	

Q.42) The "Global Alliance to Eliminate Lead Paint" is established by:

a) United Nations Environment Programme (UNEP) and the World Health Organization (WHO)

b) United Nations Development Programme (UNDP) and World Bank (WB)

c) Food and Agriculture Organization (FAO) and United Nations Environment Programme (UNEP)

d) World Bank (WB) and United Nations Environment Programme (UNEP)

100%		75%		50%		0%	

Q.37) मीथेन के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. यह एक अल्पकालिक जलवायु प्रदूषक है।

II. यह कार्बन डाइऑक्साइड के बाद वैश्विक तापन में दूसरा सबसे बड़ा योगदानकर्ता है।

III. ग्लोबल मीथेन प्लेज 2030 तक 2020 के स्तर से वैश्विक मीथेन उत्सर्जन में न्यूनतम 60% कमी का लक्ष्य रखता है।

उपर्युक्त में से कौन से कथन सही हैं?

a) केवल I और II

b) केवल II और III

c) केवल I और III

d) I, II और III

100%		75%		50%		0%	

Q.38) निम्नलिखित पदार्थों पर विचार कीजिए:

I. क्लोरोफ्लोरोकार्बन (CFCs)

II. हाइड्रोक्लोरोफ्लोरोकार्बन (HCFCs)

III. हाइड्रोफ्लोरोकार्बन (HFCs)

IV. कार्बन टेट्राक्लोराइड

उपर्युक्त में से कितने प्रमुख ओजोन क्षयकारी पदार्थ (ODS) हैं?

a) केवल एक

b) केवल दो

c) केवल तीन

d) सभी चार

100%		75%		50%		0%	

Q.39) 'LICOMK++' शब्द, जो हाल ही में समाचारों में था, निम्नलिखित में से किसके रूप में सबसे उपयुक्त रूप से वर्णित किया जाता है?

a) हिमालयी क्षेत्र में मौसम पूर्वानुमान के लिए एक उच्च-प्रदर्शन सुपरकंप्यूटिंग क्लस्टर

b) महासागर गतिकी के अध्ययन के लिए प्रयुक्त एक उच्च-रिज़ॉल्यूशन वैश्विक महासागर सिमुलेशन प्रणाली

c) उष्णकटिबंधीय क्षेत्रों में मानसून परिवर्तनशीलता की भविष्यवाणी के लिए एक एआई-आधारित प्लेटफॉर्म

d) महासागर तल के मानचित्रण के लिए डिज़ाइन किया गया एक गहरे समुद्र का अन्वेषण वाहन

100%		75%		50%		0%	

Q.40) निम्नलिखित युग्मों पर विचार कीजिए:

समाचारों में उल्लिखित संघर्ष का क्षेत्र	जिस देश में यह स्थित है
I. दीर अल-बलाह	सूडान
II. खार्तूम	सोमालिया
III. डोनेट्स्क	यूक्रेन

उपर्युक्त में से कितने युग्म सही सुमेलित हैं?

a) केवल एक

b) केवल दो

c) सभी तीन

d) कोई नहीं

100%		75%		50%		0%	

Q.41) प्रत्यक्ष वात प्रग्रहण (डायरेक्ट एयर कैप्चर), जो नई उभरती प्रौद्योगिकी है, के संदर्भ में, निम्नलिखित कथनों में से कौन-सा/कौन-से सही है/हैं?

I. इसे कार्बन प्रच्छादन की एक विधि के रूप में प्रयोग में ला सकते हैं।

II. यह प्लास्टिक उत्पादन हेतु और खाद्य प्रसंस्करण में एक बहुमूल्य उपागम हो सकता है।

III. विमानन में, यह संश्लिष्ट न्यून-कार्बन ईंधन जनन हेतु हाइड्रोजन में मिलाने के लिये कार्बन का स्रोत हो सकता है।

नीचे दिए गए कूट का उपयोग कर सही उत्तर का चयन कीजिए।

a) केवल I और II

b) केवल III

c) I, II और III

d) उपर्युक्त में से कोई भी कथन सही नहीं है

100%		75%		50%		0%	

Q.42) "ग्लोबल एलायंस टू एलिमिनेट लीड पेंट" की स्थापना किसके द्वारा की गई है?

a) संयुक्त राष्ट्र पर्यावरण कार्यक्रम (UNEP) और विश्व स्वास्थ्य संगठन (WHO)

b) संयुक्त राष्ट्र विकास कार्यक्रम (UNDP) और विश्व बैंक (WB)

c) खाद्य और कृषि संगठन (FAO) और संयुक्त राष्ट्र पर्यावरण कार्यक्रम (UNEP)

d) विश्व बैंक (WB) और संयुक्त राष्ट्र पर्यावरण कार्यक्रम (UNEP)

100%		75%		50%		0%	

Q.43) With reference to Photochemical smog, consider the following statements:

Statement I: It is generally more prevalent during the summer months compared to winter.

Statement II: It is oxidising in nature.

Statement III: The increased solar radiation and high temperatures during the summer months drive the necessary chemical reaction for its formation.

Which one of the following is correct in respect of the above statements?

a) Both Statement II and Statement III are correct and both of them explain Statement I
b) Both Statement II and Statement III are correct but only one of them explains Statement I
c) Only one of the Statements II and III is correct and that explains Statement I
d) Neither Statement II nor Statement III is correct

100%		75%		50%		0%	

Q.44) With reference to the possible effects of Acid Rain, consider the following statements:

I. The leaching of essential plant nutrients such as calcium and magnesium from the soil.
II. The destruction of beneficial nitrogen-fixing bacteria in the soil, thereby hindering plant growth.
III. Enhancing dissolved organic carbon (DOC) levels in lakes, thereby reducing UV penetration.
IV. Deterioration of exposed limestone and marble-based structures.

Which of the statements given above are correct?

a) I, II and III only
b) I and II only
c) I, II and IV only
d) I, II, III and IV

100%		75%		50%		0%	

Q.45) Which one of the following is the best description of "INS Udayagiri" and "INS Himgiri" which were seen recently in the news?

a) Nuclear-powered submarine
b) Torpedo launch and recovery vehicle
c) Coastal surveillance vessels
d) Stealth guided-missile frigates

100%		75%		50%		0%	

Q.46) How many times do the hour hand and the minute hand form a right angle in a clock between 1:00 p.m. and 5:00 p.m. (same day)?

a) 6 times
b) 7 times
c) 8 times
d) 9 times

100%		75%		50%		0%	

Q.47) The total cost of 5 proteins, 8 creatines and 10 aminos is equal to three times the total cost of 1 protein, 3 creatines and 2 aminos. In this context consider the following statements:

I. The total cost of 5 proteins, 4 creatines and 12 aminos is equal to the total cost of 3 proteins, 5 creatines and 8 aminos.
II. The total cost of 1 creatine and 4 aminos is equal to the cost of 2 proteins.

Which of the statements given above is/are correct?

a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.48) $125^4 + 25^7$ is divisible by

a) 11
b) 17
c) 65
d) 19

100%		75%		50%		0%	

Q.43) फोटोकेमिकल स्मॉग के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

कथन I: यह सामान्यतः शीत ऋतु की तुलना में ग्रीष्म ऋतु के दौरान अधिक प्रचलित होता है।

कथन II: यह ऑक्सीकारक प्रकृति का होता है।

कथन III: ग्रीष्म ऋतु के दौरान बढ़ा हुआ सौर विकिरण और उच्च तापमान इसके निर्माण के लिए आवश्यक रासायनिक अभिक्रियाओं को संचालित करते हैं।

उपर्युक्त कथनों के संदर्भ में निम्नलिखित में से कौन सा सही है?

a) कथन II और कथन III दोनों सही हैं और दोनों ही कथन I की व्याख्या करते हैं
b) कथन II और कथन III दोनों सही हैं, लेकिन उनमें से केवल एक कथन I की व्याख्या करता है
c) केवल कथन II और III में से एक सही है और वही कथन I की व्याख्या करता है
d) न तो कथन II और न ही कथन III सही है

100%		75%		50%		0%	

Q.44) अम्लीय वर्षा के संभावित प्रभावों के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. मृदा से कैल्शियम और मैग्नीशियम जैसे आवश्यक पादप पोषक तत्वों का निक्षालन।
II. मृदा में लाभकारी नाइट्रोजन स्थिरीकरण करने वाले जीवाणुओं का विनाश, जिससे पादप वृद्धि बाधित होती है।
III. झीलों में घुले हुए कार्बनिक कार्बन (DOC) के स्तर को बढ़ाना, जिससे पराबैंगनी (UV) का प्रवेश कम होता है।
IV. खुली अवस्था में स्थित चूना पत्थर और संगमरमर से निर्मित संरचनाओं का क्षरण।

उपर्युक्त में से कौन से कथन सही हैं?

a) केवल I, II और III
b) केवल I और II
c) केवल I, II और IV
d) I, II, III और IV

100%		75%		50%		0%	

Q.45) "INS उदयगिरि" और "INS हिमगिरि", जो हाल ही में समाचारों में देखे गए, निम्नलिखित में से किसका सर्वोत्तम वर्णन है?

a) परमाणु-संचालित पनडुब्बी
b) टॉरपीडो प्रक्षेपण और पुनर्प्राप्ति वाहन
c) तटीय निगरानी पोत
d) स्टील्थ निर्देशित-मिसाइल फ्रिगेट्स

100%		75%		50%		0%	

Q.46) एक ही दिन में दोपहर 1:00 बजे से 5:00 बजे के बीच घड़ी की घंटे की सुई और मिनट की सुई कितनी बार समकोण बनाती हैं?

a) 6 बार
b) 7 बार
c) 8 बार
d) 9 बार

100%		75%		50%		0%	

Q.47) 5 प्रोटीन, 8 क्रिएटिन और 10 अमीनो की कुल लागत, 1 प्रोटीन, 3 क्रिएटिन और 2 अमीनो की कुल लागत के तीन गुना के बराबर है। इस संदर्भ में निम्नलिखित कथनों पर विचार कीजिए:

I. 5 प्रोटीन, 4 क्रिएटिन और 12 अमीनो की कुल लागत, 3 प्रोटीन, 5 क्रिएटिन और 8 अमीनो की कुल लागत के बराबर है।
II. 1 क्रिएटिन और 4 अमीनो की कुल लागत, 2 प्रोटीन की लागत के बराबर है।

उपर्युक्त में से कौन-सा/से कथन सही है/हैं?

a) केवल I
b) केवल II
c) I और II दोनों
d) न तो I और न ही II

100%		75%		50%		0%	

Q.48) $125^4 + 25^7$ किससे विभाज्य है?

a) 11
b) 17
c) 65
d) 19

100%		75%		50%		0%	

Directions for the following 2 (two) items:
Read the following passage and answer the items that follow the passage. Your answers to these items should be based on the passage only

Passage - 1

India's rural economy continues to be shaped by the uncertainties of water availability. While large irrigation projects often dominate public discourse, nearly 52 per cent of the country's gross cropped area remains rainfed. In this context, watershed development deserves renewed attention as a central pillar of India's strategy for sustainable rural growth. Structures such as check dams, farm ponds, contour bunds and percolation tanks have been associated with rises in groundwater levels, expansion of irrigated area and increases in cropping intensity often ranging from 10-20 percentage points. Such improvements allow farmers to diversify into horticulture, pulses and other higher value crops. The economic impact of watershed development extends beyond crop output. Improved water availability often stimulates investments in livestock, agro processing and small rural enterprises. Watershed programmes also interact closely with rural infrastructure development. Increased water security encourages the creation of farm level assets, minor irrigation channels and storage facilities. Taken together, these outcomes demonstrate that managing water at the micro catchment level can generate macro economic dividends.

Q.49) Which one of the following statements best reflects the most logical and rational inference that can be drawn from the passage?

a) Watershed development contributes to increased water availability by groundwater recharge mechanisms.
b) Local level interventions are needed to resolve the problem of water scarcity in rural India.
c) Efficient micro-level water management can generate tangible and sustained economic gains for the rural economy.
d) Watershed development reduces the dependence of agricultural output on uncertain rainfall.

100%		75%		50%		0%	

Q.50) With reference to the above passage, the following assumptions have been made:

I. Rural India is experiencing severe water scarcity due to declining groundwater levels.
II. Large scale irrigation projects cannot solve the problem of water scarcity in rural India.

Which of the above assumptions is/are valid?

a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

निर्देश (अगले 2 प्रश्नों के लिए):
निम्नलिखित परिच्छेद को पढ़िए और उसके बाद दिए गए प्रश्नों के उत्तर दीजिए। इन प्रश्नों के उत्तर केवल परिच्छेद के आधार पर होने चाहिए।

परिच्छेद - 1

भारत की ग्रामीण अर्थव्यवस्था जल उपलब्धता की अनिश्चितताओं द्वारा निरंतर प्रभावित होती रहती है। जबकि वृहत सिंचाई परियोजनाएँ अक्सर सार्वजनिक विमर्श में प्रमुख स्थान रखती हैं, देश के कुल बोए गए क्षेत्र का लगभग 52 प्रतिशत अभी भी वर्षा-आधारित है। इस संदर्भ में, जलग्रहण क्षेत्र विकास को भारत की सतत ग्रामीण विकास रणनीति के एक केंद्रीय स्तंभ के रूप में पुनः महत्व दिया जाना चाहिए। चेक डैम, फार्म पॉन्ड, कंटूर बंड्स और पर्कोलेशन टैंक जैसी संरचनाएँ भूजल स्तर में वृद्धि, सिंचित क्षेत्र के विस्तार और फसल गहनता में वृद्धि (अक्सर 10-20 प्रतिशत अंक तक) से जुड़ी रही हैं। ऐसे सुधार किसानों को बागवानी, दलहन और अन्य उच्च मूल्य वाली फसलों की ओर विविधीकरण करने में सक्षम बनाते हैं। जलग्रहण क्षेत्र विकास का आर्थिक प्रभाव केवल फसल उत्पादन तक सीमित नहीं है। बेहतर जल उपलब्धता अक्सर पशुपालन, कृषि प्रसंस्करण और छोटे ग्रामीण उद्यमों में निवेश को प्रोत्साहित करती है। जलग्रहण कार्यक्रम ग्रामीण अवसंरचना विकास के साथ भी निकटता से जुड़े होते हैं। जल सुरक्षा में वृद्धि कृषक स्तर पर परिसंपत्तियों, लघु सिंचाई नहरों तथा भंडारण सुविधाओं के निर्माण को प्रोत्साहित करती है। समग्र रूप से, ये परिणाम दर्शाते हैं कि सूक्ष्म जलग्रहण स्तर पर जल प्रबंधन व्यापक आर्थिक लाभ उत्पन्न कर सकता है।

Q.49) निम्नलिखित में से कौन-सा कथन परिच्छेद से निकाला जा सकने वाला सबसे तार्किक और युक्तिसंगत निष्कर्ष प्रस्तुत करता है?

a) जलग्रहण क्षेत्र विकास, भूजल पुनर्भरण तंत्र के माध्यम से जल उपलब्धता में वृद्धि करता है।
b) ग्रामीण भारत में जल की कमी की समस्या के समाधान के लिए स्थानीय स्तर के हस्तक्षेपों की आवश्यकता है।
c) सूक्ष्म-स्तर पर कुशल जल प्रबंधन ग्रामीण अर्थव्यवस्था के लिए ठोस और सतत आर्थिक लाभ उत्पन्न कर सकता है।
d) जलग्रहण क्षेत्र विकास, कृषि उत्पादन की अनिश्चित वर्षा पर निर्भरता को कम करता है।

100%		75%		50%		0%	

Q.50) उपर्युक्त परिच्छेद के संदर्भ में, निम्नलिखित धारणाएं बनाई गई हैं:

I. ग्रामीण भारत में घटते भूजल स्तर के कारण गंभीर जल संकट का सामना करना पड़ रहा है।
II. बड़े पैमाने की सिंचाई परियोजनाएं ग्रामीण भारत में जल संकट की समस्या का समाधान नहीं कर सकती हैं।

उपर्युक्त में से कौन-सी धारणा/धारणाएं सही है/हैं?

a) केवल I
b) केवल II
c) I और II दोनों
d) न तो I और न ही II

100%		75%		50%		0%	

SPACE FOR ROUGH WORK

Q.1) Consider the following statements:
Statement I: Studies indicate that carbon dioxide emissions from cement industry account for more than 5% of global carbon emissions.
Statement II: Silica-bearing clay is mixed with limestone while manufacturing cement.
Statement III: Limestone is converted into lime during clinker production for cement manufacturing.
Which one of the following is correct in respect of the above statements?
a) Both Statement II and Statement III are correct and both of them explain Statement I
b) Both Statement II and Statement III are correct but only one of them explains Statement I
c) Only one of the Statements II and III is correct and that explains Statement I
d) Neither Statement II nor Statement III is correct

Ans) b
Exp) Option b is the correct answer
The cement industry is a significant source of carbon dioxide emissions due to both the chemical reactions involved in producing clinker (the main ingredient of cement) and the burning of fossil fuels to power the production process. Specifically, calcining limestone (a primary ingredient) to produce lime (CaO) releases CO2, and burning fossil fuels to heat the kilns also contributes to emissions.
Statement I is correct: Global cement manufacturing produced 1.6 billion metric tonnes of CO2 in 2022, the latest year for which there are figures – that's about 8% of the world's total CO2 emissions.
Statement II is correct: Silica-bearing clay is mixed with limestone while manufacturing cement. Cement production involves mixing limestone with other materials like clay, which contains silica, along with other minerals. These materials are then processed into clinker, a key component of cement.
Statement III is correct: In the process of making cement clinker, limestone, which is primarily calcium carbonate (CaCO3), is heated to high temperatures (1450°C or 2640°F) in a kiln. This process, called calcination, decomposes the limestone into calcium oxide (CaO), commonly known as lime, and carbon dioxide (CO2). The lime then reacts with other raw materials like clay and silica to form the clinker, which is the main ingredient in cement.
Both Statement II and Statement III are correct but only one of them (i.e. Statement III) explains Statement I:

- Statement I discusses the environmental impact (carbon emissions) of the cement industry.
- Statement III explains why the cement industry has significant carbon emissions (as mentioned in Statement I), due to the conversion of limestone to lime releasing CO2.

Source: UPSC CSE Pre 2025
Subject:) Environment
Topic:) Environmental Pollution
Subtopic:)

Q.2) Which of the following is/are anthropogenic sources of mercury pollution?
I. Artisanal and small-scale gold mining
II. Volcanic eruptions
III. Coal-fired power plants
IV. Burning of municipal and medical waste
Select the correct answer using the code given below.
a) I and III only
b) II only
c) I, III, and IV only
d) I, II, III and IV

Ans) c
Exp) Option c is the correct answer.
Option I is correct: Artisanal and small-scale gold mining: Artisanal and small-scale gold mining releases mercury during the amalgamation process. In the process, mercury is used to extract gold and later vaporized into the atmosphere.

As per report by UNEP, Artisanal mining accounts for 20 per cent of the global supply of gold and generates approximately US$30 billion annually. Artisanal and small-scale gold mining emits over 2,000 tonnes of mercury per year. This includes emissions to air from heating amalgams, as well as direct losses of mercury to land and water

Option II is incorrect: Volcanic eruptions: Volcanic eruptions are natural sources of mercury, releasing it from the Earth's crust into the atmosphere along with other gases and particulates. It is not an anthropogenic source.

Option III is correct: Coal-fired power plants: Coal-fired power plants emit mercury as coal contains trace amounts of mercury, which is released during combustion into the atmosphere.

Option IV is correct: Waste incineration: Many consumer products (batteries, thermometers, switches) and medical devices contain mercury. When waste is incinerated scrubbing technology, the mercury is released.

Source:) https://www.unep.org/globalmercurypartnership/what-we-do/artisanal-and-small-scale-gold-mining-asgm

https://www.who.int/news-room/fact-sheets/detail/mercury-and-health

Subject:) Environment

Topic:) Environmental Pollution

Subtopic:)

Q.3) With reference to microplastics, consider the following statements:

I. These are defined as shreds of plastic less than 5 millimeters in length

II. All microplastics presently come from the slow disintegration of larger plastic products.

III. Recent studies, in humans, showed that microplastics can cross placental barriers, which would disrupt the normal fetal growth and development.

IV. As per the recent WHO guidelines, in drinking water, microplastics up to 10,000 particles/L are considered as safe for drinking.

Which of the statements given above are correct?

a) I and III only

b) II, III and IV

c) I, II and III

d) I, II and IV

Ans) a

Exp) Option a is the correct answer

Statement I is correct: Microplastics though encompass a wide range of materials composed of different substances, with different densities, chemical compositions, shapes and sizes. **Microplastics are generally defined as shreds of plastic less than 5 millimeters in length.**

Statement II is incorrect: Microplastics are either manufactured – for instance, microbeads that are used in cosmetics and beauty products – or they are formed when larger pieces of plastic break down. The small, shiny particles advertised as cooling crystals in certain toothpastes qualify as microplastics if the ingredients of the toothpaste mention polyethylene.

Thus, here are two main types of microplastics. Some plastics are made to be small. These are known as primary microplastics, like the microbeads intentionally added to face washes and other personal care products. But most microplastics come from the slow disintegration of larger plastic products and are known as secondary microplastics.

Statement III is correct: As per recent studies, microplastics may even cross the placental barrier, which is typically designed to protect the developing fetus from harmful substances. This penetration raises concerns about the possible effects of microplastics on fetal development and long-term health. **This can disrupt normal fetal growth and development.**

Statement IV is incorrect: There is no 'acceptable' level of microplastics. As per the WHO in freshwater studies, reported microplastic particle counts ranged from around 0 to 1000 particles/L. **WHO has not set any limit with respect to microplastics in drinking water and rearch for this purpose is still going on.**

Source: https://indianexpress.com/article/explained/explained-what-are-microplastics-the-new-pollutant-you-are-breathing-5913372/

https://timesofindia.indiatimes.com/life-style/health-fitness/health-news/microplastics-in-the-womb-what-we-know-about-risks-to-unborn-babies/articleshow/123678825.cms
https://cdn.who.int/media/docs/default-source/wash-documents/microplastics-in-dw-information-sheet190822.pdf
Subject:) Environment
Topic:) Environmental Pollution
Subtopic:)

Q.4) 'Carbosulfan' and 'Fenthion', which were seen recently in the news, are:
a) Chemical pesticides/insecticides added to Annex III of the Rotterdam Convention.
b) Prohibited chemical preservatives in processed foods under the Food Safety and Standards Authority of India regulations.
c) Primary chemical markers to identify the geographical origin of Basmati rice.
d) Bio-fertilizers approved in India for promotion of organic farming.

Ans) a
Exp) Option a is the correct answer.
Carbosulfan and Fenthion are toxic chemical pesticides/insecticides which have hazardous effects on human health and the environment. They have been added in Annex III of the Rotterdam Convention, which enables prior informed consent (PIC) procedures to regulate their international trade. Annex III of the Rotterdam Convention is a list of hazardous chemicals that are subject to the Prior Informed Consent (PIC) procedure. Once a chemical is added to Annex III, any country exporting that chemical must first get the "informed consent" of the importing country.
Source:) https://www.downtoearth.org.in/governance/rotterdam-convention-deadlocked-on-chlorpyrifos-as-countries-resist-transparency-in-hazardous-pesticide-trade
https://sansad.in/getFile/loksabhaquestions/annex/14/AU4136.pdf?source=pqals
https://www.pic.int/Implementation/PublicAwareness/NewsFeatures/ListingofcarbosulfanandfenthioninAnnexIII/tabid/10381/language/en-US/Default.aspx#:~:text=Postal%20address:%20Avenue%20de%20la,10381
Subject:) Environment
Topic:) Environmental Pollution
Subtopic:)

Q.5) In the context of thermal pollution, as the temperature of a water body increases, the pH value of the water will generally:
a) Increase
b) Decrease
c) Remain unchanged
d) Increases initially and then decreases

Ans) b
Exp) Option b is the correct answer.
Thermal pollution occurs when industries discharge heated water into natural water bodies such as rivers and lakes. This rise in temperature affects not only dissolved oxygen levels but also the chemical properties of water, including its pH. One key concept is that the **self-ionization of water ($H_2O \rightleftharpoons H^+ + OH^-$)** is an **endothermic process**, meaning it is favored at higher temperatures.**As temperature rises,** the ionization of water increases, producing more H^+ ions. This results in a **lower pH value**. For example, while neutral pH is 7 at 25°C, it decreases to around **6.15 at 100°C**,
Source:) https://egyankosh.ac.in/bitstream/123456789/7568/1/Unit-7.pdf page 104
Subject:) Environment
Topic:) Environmental Pollution
Subtopic:)

Q.6) Consider the following statements:
Statement I: Phthalates are mainly used as plasticizers to increase the durability of plastics.
Statement II: Phthalates form permanent chemical bonds with polymer chains of plastic.
Which one of the following is correct in respect of the above statements?
a) Both Statement I and Statement II are correct and Statement II explains Statement I.
b) Both Statement I and Statement II are correct but Statement II does not explain Statement I.
c) Statement I is correct but Statement II is not correct.
d) Statement I is not correct but Statement II is correct.

Ans) c
Exp) Option c is the correct answer.
Statement I is correct: Phthalates are mainly used as plasticizers. Plasticizers are the substances added to plastics to increase their flexibility, transparency, durability, and longevity. In a rigid polymer like Polyvinyl Chloride (PVC), the polymer chains are tightly packed. **Phthalates insert themselves between these long chains, acting like a "lubricant" that allows the chains to slide past each other. It makes PVC softer and more workable for applications like cables, flooring, and medical devices.**
Statement II is incorrect: Phthalates are not chemically bonded to polymer chains. They are physically mixed and can migrate out over time. This non-permanent bonding is the reason for plastics to lose flexibility and release phthalates into the environment.
Hence, **Statement I is correct, but Statement II is not correct.**
Source:) https://timesofindia.indiatimes.com/city/patna/study-reveals-highlevel-of-phthalatesin-parts-of-city-soil/articleshow/100753997.cms https://egyankosh.ac.in/bitstream/123456789/31850/1/Unit-4.pdf Pg: 11 https://indianexpress.com/article/lifestyle/food-wine/phthalates-detergent-gloves-mcdonalds-burger-king-pizza-hut-foods-study-7596624/
https://www.downtoearth.org.in/waste/plastic-s-toxic-trail-what-are-these-polymers-and-what-do-they-contain--95792
Subject:) Environment
Topic:) Environmental Pollution
Subtopic:)

Q.7) With reference to the World Air Quality Report 2024–25, consider the following statements:
I. Delhi has been ranked as the most polluted national capital in the world in terms of air quality.
II. Byrnihat city in Meghalaya is the cleanest city in India in terms of air quality.
III. The report uses PM2.5 as the primary metric for air quality reporting.
Which of the statements given above are correct?
a) I and II only
b) II and III only
c) I and III only
d) I, II and III

Ans) c
Exp) Option c is the correct answer.
The World Air Quality Report is **released by the Swiss organization, IQAir** which provides a comprehensive look at global air pollution trends.
Statement I is correct: The World Air Quality Report 2024–25 recorded an annual average PM2.5 concentration for Delhi that is approximately 18 times higher than the World Health Organization (WHO) annual safety guideline. For the seventh consecutive year, New Delhi has been ranked as the most polluted capital city in the world.
Statement II is incorrect: Byrnihat, located on the Assam-Meghalaya border, was ranked as the most polluted city in India. It is the most polluted city in India due to industrial and transport emissions.
The annual average PM2.5 concentration in Byrnihat was 128.2 micrograms per cubic m (µg/m3), many times the World Health Organization's annual air quality guideline of 5 µg/m3.

Statement III is correct: The report uses PM2.5 concentration as the key indicator for assessing air quality. PM2.5 is considered the most hazardous air pollutant to human health because its tiny size allows it to bypass the nose and throat, penetrating deep into the lungs and even entering the bloodstream.
Source:) https://timesofindia.indiatimes.com/city/delhi/for-7th-year-delhi-retains-title-of-most-polluted-capital-in-world/articleshow/118912400.cms#:~:text=NEW%20DELHI%3A%20New%20Delhi%20retained,With%20an%20annual%20PM2.
https://www.iqair.com/in-en/world-air-quality-report https://indianexpress.com/article/explained/explained-climate/why-byrnihat-on-assam-meghalaya-border-has-worlds-worst-air-9883680/
Subject:) Environment
Topic:) Environmental Pollution
Subtopic:)

Q.8) In the context of India, consider the following statements:
Statement I: As per National Ambient Air Quality Standards, unlike some other pollutants, tropospheric ozone levels are monitored in an eight-hour or one-hour average period, instead of annual or 24-hour averages.
Statement II: Tropospheric Ozone is highly reactive.
Which one of the following is correct in respect of the above statements?
a) Both Statement I and Statement II are correct and Statement II explains Statement I.
b) Both Statement I and Statement II are correct but Statement II does not explain Statement I.
c) Statement I is correct but Statement II is incorrect.
d) Statement I is incorrect but Statement II is correct.

Ans) a
Exp) Option a is the correct answer
Statement I is correct: The ambient air quality standards for Ozone (O3) i.e. Tropospheric Ozone, have been notified by the Central Pollution Control Board (CPCB) as one of the twelve (12) pollutants under the National Ambient Air Quality Standard (NAAQS), 2009. The ambient air quality standards for Ozone (O3) is prescribed as 100 µg/m3 for 8-hourly monitored value and 180 µg/m3 for 1-hourly monitored value for industrial, residential, rural and ecologically sensitive areas. **The time-weighted average level of Tropospheric Ozone is 8 hours or 1 hour unlike 24 hours or annual average for other pollutants.** Also, carbon monoxide level is monitored in an **8 hours or 1 hour average period** As per the NAAQS, the method of measurement of Ozone (O3) in ambient air is UV photometric, Chemiluminescence, and Chemical method.
Statement II is correct: Tropospheric (or ground-level) ozone is a short-lived climate pollutant that remains in the atmosphere for only hours to weeks. **Tropospheric ozone is a highly reactive oxidant.** It is highly reactive because it contains weak oxygen bonds that easily break in the presence of sunlight, releasing a single, highly unstable oxygen atom.
Statement II explains Statement I: Due to its reactive nature, O_3 levels are monitored in an eight-hour average period, instead of 24-hour averages, unlike other pollutants. The prescribed ozone standard for an eight-hour period is 100 micrograms/cubic metre as per the National Ambient Air Quality Standards (NAAQ).
Source: https://indianexpress.com/article/india/summer-time-ozone-level-megacities-cse-study-10130383/
https://www.ccacoalition.org/short-lived-climate-pollutants/tropospheric-ozone
Subject:) Environment
Topic:) Environmental Pollution
Subtopic:)

Q.9) The National Level Pollution Response Exercise (NATPOLREX) is conducted by:
a) Central Pollution Control Board.
b) National Environmental Engineering Research Institute (NEERI).
c) Indian Coast Guard.
d) Ministry of Environment, Forest and Climate Change.

Ans) c

Exp) Option c is the correct answer.

The National Level Pollution Response Exercise (NATPOLREX) is a major biennial exercise conducted by the Indian Coast Guard (ICG). The primary objective of the exercise is to evaluate the preparedness and response capabilities of the Indian Coast Guard and other stakeholders in handling large-scale marine oil spills and chemical pollution in Indian waters.

Source:) https://www.pib.gov.in/PressReleasePage.aspx?PRID=2175452&reg=3&lang=2
https://indianexpress.com/article/cities/ahmedabad/coast-guard-dg-to-oversee-sea-exercises-carried-out-off-vadinar-coast-on-nov-25-9036904/

Subject:) Environment

Topic:) Disaster Management

Subtopic:)

Q.10) With reference to the Wassenaar Arrangement, consider the following statements:

I. It is a multilateral export control regime headquartered in Hague, Netherlands.

II. It promotes transparency and greater responsibility in the transfer of conventional arms and dual-use goods and technologies between member countries.

III. India is a member of the Arrangement.

Which of the statements given above is/are correct?

a) I and II

b) II and III

c) I and III

d) II only

Ans) b

Exp) Option b is the correct answer.

Statement I is incorrect: Wassenaar Arrangement is a multilateral export control regime, established in 1996 and headquartered in Vienna, Austria. The arrangement succeeded the Cold War-era Coordinating Committee for Multilateral Export Controls (COCOM).

Statement II is correct: The arrangement promotes transparency and greater responsibility in the transfer of conventional arms and dual-use goods and technologies between member countries. Its broader objective is to prevent any exports or activities that could threaten or destabilize international peace and security.

Statement III is correct: The arrangement currently has 42 member countries, with India being the most recent to join in 2017. Following its admission, India integrated the arrangement's control lists into its Special Chemicals, Organisms, Materials, Equipment, and Technologies (SCOMET) framework.

Source: SFG Essential- Current Affairs- Quarterly Compilation (July–September 2025- Page no: 44

Subject:) Current Affairs

Topic:) Security

Subtopic:)

Q.11) Consider the following statements:

1. Agricultural soils release nitrogen oxides into the environment.
2. Cattle release ammonia into the environment.
3. Poultry industry releases reactive nitrogen compounds into environment.

Which of the statements given above is/are correct?

a) 1 and 3 only

b) 2 and 3 only

c) 2 only

d) 1, 2 and 3

Ans) d

Exp) Option d is the correct answer.

Statement 1 is correct: Agricultural soils release nitrogen oxides (NOx) into the environment through the process of nitrification and denitrification, which convert nitrogen in fertilizers and manure into nitrate and nitrous oxide. Nitrous oxide is a powerful greenhouse gas that contributes to global warming and ozone depletion.

Statement 2 is correct: Cattle release ammonia (NH3) into the environment through their urine and feces. Ammonia is a common by-product of animal waste due to the often-inefficient conversion of feed nitrogen into animal product. Ammonia is harmful to the environment because it contributes to air pollution, acid rain, eutrophication, and biodiversity loss.

Statement 3 is correct: Poultry industry releases reactive nitrogen compounds (RNCs) such as ammonia, nitrous oxide, and nitrogen dioxide into the environment through the use of synthetic fertilizers, manure management, and poultry feed. These compounds are released through the use of nitrogen-rich fertilizers and manure management practices. The release of these compounds contributes to air pollution and can cause respiratory problems.

Source: UPSC CSE Pre 2019
Subject:) Environment
Topic:) Various Environmental Issues
Subtopic:)

Q.12) With reference to biomining and biocapping, consider the following statements:
I. Biomining involves using microorganisms to degrade waste and recover useful materials.
II. Biocapping involves covering waste dumps with layers of soil to prevent the release of pollutants.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

Ans) c
Exp) Option c is the correct answer.

Statement I is correct: Biomining is a scientific method used to remediate "legacy waste" (old, accumulated garbage) in landfills. It involves **using microorganisms to break down and stabilize the organic components of the waste.** After stabilization, the waste is **mechanically screened to recover useful materials** such as recyclables (plastic, glass) and bio-earth (soil-like material), which helps in reclaiming the land.

Statement II is correct: Biocapping is a containment strategy used for the scientific closure of waste dumps. It involves covering the waste with multiple layers of soil, clay, or synthetic liners. The goal is to prevent water from entering the waste (reducing leachate) and to stop the release of pollutants.

Source:) https://www.thehindu.com/news/cities/Coimbatore/coimbatore-corporation-plans-bio-mining-to-reclaim-land-at-kavundampalayam-compost-yard/article70139529.ece#:~:text=The%20Coimbatore%20Corporation%20plans%20to,bio%2Dcapping%20for%20nearly%20a%20decade.
https://www.pib.gov.in/PressReleseDetailm.aspx?PRID=2221171&reg=3&lang=2
https://cpcb.nic.in/uploads/LegacyWasteBiomining_guidelines_29.04.2019.pdf (page-10)
Subject:) Environment
Topic:) Various Environmental Issues
Subtopic:)

Q.13) The Global Outlook Council on Water Investments is an initiative associated with:
a) International Water Management Institute
b) World Bank
c) G20
d) Wetland International

Ans) c
Exp) Option c is the correct answer.

The Global Outlook Council on Water Investments is an initiative of the G20. It serves as a high-level platform designed to mobilize both public and private capital for the water sector. The council focuses on developing sustainable financing models and promoting international cooperation to bridge the global investment gap in water-related infrastructure.
Source:) https://economictimes.indiatimes.com/news/india/south-african-president-ramaphosa-welcomes-modi-other-leaders-to-global-outlook-council-on-water-investments/articleshow/123300490.cms?from=mdr
Subject:) Environment
Topic:) Various Environmental Issues
Subtopic:)

Q.14) With reference to the Environment Protection (Management of Contaminated Sites) Rules, 2025, consider the following statements:
I. These rules have been notified under the Air (Prevention and Control of Pollution) Act, 1981
II. The rules provide a framework to identify and remediate chemically contaminated sites.
III. These rules exclude contamination of sites arising from radioactive waste.
Which of the statements given above are correct?
a) I and II only
b) II and III only
c) I, II and III
d) I and III only

Ans) b
Exp) Option b is the correct answer.
Statement I is incorrect: The Ministry of Environment, Forest and Climate Change notified the Environment Protection (Management of Contaminated Sites) Rules, 2025 under the Environment (Protection) Act, 1986.
Statement II is correct. The Environment Protection (Management of Contaminated Sites) Rules, 2025, were notified to provide a comprehensive **regulatory framework for the identification, investigation, and remediation of sites** where soil or groundwater has been contaminated by hazardous substances.
Statement III is correct. These rules **specifically exclude contamination of sites arising from radioactive waste,** which continues to be governed under the Atomic Energy Act, 1962.
As per the rules, contamination resulting from radioactive waste, mining operations, pollution of the sea by oil, solid waste from dump sites would not be dealt with under the provisions of these laws as they are governed by separate legislation.
Source:) https://www.pib.gov.in/PressReleasePage.aspx?PRID=2157518&reg=3&lang=2
https://www.thehindu.com/sci-tech/energy-and-environment/procedure-to-redress-chemically-contaminated-sites-gets-legal-teeth/article69865893.ece
Subject:) Environment
Topic:) Various Environmental Issues
Subtopic:)

Q.15) With reference to the Armed Forces (Special Powers) Act, 1958, which of the following authorities has/have the power to declare a region as "disturbed area"?
I. Supreme Court of India
II. Central Government
III. Governor of the concerned State
IV. High Court of the concerned State
Select the correct answer using the code given below:
a) III only
b) III and IV
c) II and III
d) I and II

Ans) c

Exp) Option c is the correct answer.

Armed Forces (Special Powers) Act, 1958 is a law passed by Parliament granting the armed forces special powers and legal immunity to maintain public order in designated "disturbed areas".

A disturbed area is one officially notified under Section 3 of AFSPA. Any whole or part of a state where conditions have become dangerous or disturbed, necessitating the deployment of armed forces to assist civil authorities, can be declared as such.

The power to declare a disturbed area lies with the Central Government, or the Governor of the concerned State, or the administrator of the Union Territory (but not with High Courts of concerned States or the Supreme Court of India).

Source: SFG Essential- Current Affairs- Quarterly Compilation (July–September 2025- Page no: 44

Subject:) Current Affairs

Topic:) Polity

Subtopic:)

Q.16) Dissolved Oxygen (DO) in water bodies is influenced by various physical and environmental factors. In this context, consider the following statements:

I. Under similar conditions, cold water can hold more dissolved oxygen than warm water.

II. Under similar temperature and pressure conditions, freshwater can hold more dissolved oxygen than seawater.

III. Compared to lower latitudes, dissolved oxygen levels are generally higher at high altitudes due to lower temperatures.

IV. Stagnant water bodies tend to have lower dissolved oxygen compared to flowing or turbulent water bodies.

Which of the statements given above are correct?

a) I, II and III only

b) I, II and IV only

c) III and IV only

d) I, II, III and IV

Ans) b

Exp) Option b is the correct answer.

Dissolved Oxygen (DO) refers to the amount of oxygen gas present in water, typically measured in mg/L. Although only a small quantity of oxygen (about a few parts per million) dissolves in water, it is **crucial for the survival of aquatic organisms** such as fish, plankton, and benthic fauna. The level of DO in water bodies is controlled by several physical and biological factors such as **temperature, salinity, atmospheric pressure, water movement, and biological activity**.

Statement I is correct. The solubility of gases, including oxygen, **decreases with increasing temperature**. For example, at 0°C, water can hold around 14–15 mg/L of oxygen, whereas at 35°C, it drops to about 7 mg/L. Thus, colder water bodies such as mountain streams tend to have higher DO levels, which supports richer aquatic life.

Statement II is correct. The presence of salts in seawater slightly reduces the solubility of oxygen. Freshwater can hold **more dissolved oxygen than saline water** under similar conditions. For instance, at 0°C, freshwater holds about 14–15 ppm DO, while seawater holds only about 10–11 ppm.

Statement III is incorrect. While lower temperatures at high altitudes favor higher oxygen solubility, **atmospheric (barometric) pressure decreases with altitude**, which reduces the overall solubility of oxygen in water. Hence, despite colder conditions, DO levels are generally **lower at high altitudes** compared to sea level.

Statement IV is correct. Flowing or turbulent water (like rivers and streams) promotes **aeration**, allowing oxygen from the atmosphere to dissolve into water efficiently. In contrast, stagnant water bodies (ponds, lakes) have limited mixing. Additionally, decomposition of organic matter by bacteria consumes oxygen, further reducing DO levels, especially during summer when oxygen demand is high.

Source:) https://www.egyankosh.ac.in/bitstream/123456789/71132/1/Unit-3.pdf page 49

https://egyankosh.ac.in/bitstream/123456789/70784/1/Unit-7.pdf page 26

https://egyankosh.ac.in/bitstream/123456789/79935/1/Unit-6.pdf page 168, 193

Subject:) Environment

Topic:) Various Environmental Issues
Subtopic:)

Q.17) Recently, the government carried out reforms in the Disaster Management framework through the enactment of the Disaster Management (Amendment) Act, 2025. In this context, consider the following statements:
I. It empowers the state government to constitute Urban Disaster Management Authority (UDMA) in the state capital.
II. It shifts the responsibility of preparing national disaster management plans from the National Disaster Management Authority (NDMA) to the National Executive Committee.
III. It mandates the creation of a National Disaster Database incorporating risk assessments, mitigation strategies, and real-time disaster data.
Which of the statements given above are correct?
a) I, II and III
b) I and III only
c) II and III only
d) I and II only

Ans) b
Exp) Option b is the correct answer.
Statement I is correct. The Amendment Act inserts a provision (Section 41A) empowering State Governments to constitute **Urban Disaster Management Authorities (UDMAs)** in State Capitals and all cities having Municipal Corporation (excluding NCT of Delhi and Union Territory of Chandigarh). These bodies are intended to address **urban-specific disaster risks** such as flooding, heatwaves, and infrastructure stress, which are becoming more prominent due to rapid urbanisation.
Statement II is incorrect. The Amendment Act **strengthens the role of the National Disaster Management Authority (NDMA)** by assigning it the responsibility of preparing national disaster management plans. Earlier, this function was largely performed by the National Executive Committee (NEC).
Statement III is correct. The Amendment mandates the creation of a **National Disaster Database**, which includes **risk assessments, mitigation plans, and real-time disaster data**. This is a significant step towards **technology-driven and evidence-based disaster management**, improving preparedness, forecasting, and response.
Source: https://www.pib.gov.in/PressReleasePage.aspx?PRID=2226438&reg=3&lang=1
https://www.newsonair.gov.in/rajya-sabha-debates-disaster-management-bill-2024-amid-opposition-criticism/
Subject:) Environment
Topic:) Disaster Management
Subtopic:)

Q.18) With reference to pollution monitoring in India, consider the following information:

Index	**Implementing agency**	**Description**
I.National Air Quality Monitoring Programme (NAMP)	Central Pollution Control Board	It covers all the districts of India
II. National Air Quality Index (NAQI)	Commission for Air Quality Management in the National Capital Region and Adjoining Areas	It communicates air quality in simple terms through One Number – One Color – One Description.
III. System of Air Quality Forecasting and Research (SAFAR)	Indian Institute of Tropical Meteorology (IITM)	Apart from forecasting air quality it also provides forecasts on weather, emissions and UV radiation.

In which of the above rows is the given information correctly matched?
a) Only one

b) Only two
c) All the three
d) None

Ans) a
Exp) Option a is the correct answer.
Row I is incorrect. The **Central Pollution Control Board** is executing a nation-wide programme of ambient air quality monitoring known as **National Air Quality Monitoring Programme (NAMP)**. Under NAMP, four air pollutants, viz, **Sulphur Dioxide (SO2), Oxides of Nitrogen as NO2, Respirable Suspended Particulate Matter (RSPM / PM10) and Fine Particulate Matter (PM2.5)** have been identified for regular monitoring at 966 operating stations in **419 cities/towns** in 28 states and 7 Union Territories. It d**oes not cover all the district**s of India.
Row II is incorrect. The AQI in India is **developed and implemented by the Central Pollution Control Board (CPCB)**, not by the Commission for Air Quality Management (CAQM). It is a tool to communicate real-time air quality in simple terms–"One Number – One Color – One Description". It ranks air quality from 0–500 across six categories, ranging from "Good" to "Severe," to indicate potential health impacts based on eight key pollutants (PM10, PM2.5, NO2, SO2, CO, O3, NH3, and Pb).
Row III is correct. SAFAR is implemented under the MoES but is implemented by **IITM,** an autonomous institution of the MoES based in Pune, and the I**ndian Meteorological Department (IMD)**. It provides real-time air quality data and forecasts, and monitors a wide range of pollutants including PM1, PM10, PM2.5, O3, NO2, NO, NOx, CO, Organic Carbon (OC) and Black Carbon (BC), Hg, Benzene, Toulene, Xylene. It provides forecasting information on air quality and not simply monitors the quality of air. **It now also provides forecasts on weather, emissions and UV radiation.**
Source:) https://cpcb.nic.in/about-namp/
https://cpcb.nic.in/displaypdf.php?id=bmF0aW9uYWwtYWlyLXF1YWxpdHktaW5kZXgvQWJvdXRfQVFJLnBkZg==
https://www.indiascienceandtechnology.gov.in/node/111168
http://164.100.94.191/niti/writereaddata/files/bestpractices/SAFAR%20System%20of%20Air%20quality%20Forecasting%20And%20Research%20in%20metropolitan%20cities%20like%20Delhi.pdf
Subject:) Environment
Topic:) Environmental Pollution
Subtopic:)

Q.19) In the context of Environmental Impact Assessment (EIA) rules in India, consider the following projects/activities:
I. Starting a petroleum refining project
II. Construction of highway projects of strategic importance, which are within 100 km from the Line of Control.
III. Development of a coal-based thermal power plant with a capacity of 600 MW.
Which of the above projects are most likely to require prior environmental clearance?
a) I and III only
b) I and II only
c) II and III only
d) I, II and III

Ans) a
Exp) Option a is the correct answer.
Environmental Impact Assessment (EIA) is a statutory process under the Environment (Protection) Act, 1986, used to evaluate the likely environmental consequences of proposed projects before granting approval. Under the EIA Notification, 2006 (as amended), certain categories of projects mandatorily require prior Environmental Clearance (EC)
Statement I is correct. As per the EIA framework, **the petroleum refining industry requires mandated Environmental clearance.**
Statement II is incorrect. As per recent amendments to EIA rules, **highway projects of strategic and defence importance within 100 km of the Line of Control (LoC)** are specifically exempted from the requirement of prior EC. This exemption is granted to facilitate faster execution of critical infrastructure projects in sensitive border areas.

Statement III is correct. Thermal power plants a**bove 500 MW** capacity for coal/lignite-based plants are classified under Category A and mandatorily require **prior Environmental Clearance.** Thus, a **coal-based thermal plant with 600 MW capacity** will require **prior environmental clearance** from the appropriate authority.
Source: https://indianexpress.com/article/india/environment-ministry-amends-eia-rules-exempts-highways-strategic-importance-8039740/
https://parivesh.nic.in/utildoc/services/notification/ec/EIA_Notification_2006_including_all_amendments_till_January_2026_v1.pdf page 18-20
Subject:) Environment
Topic:) Environmental Impact Assessment
Subtopic:)

Q.20) Which of the following best describes the objective of "Operation Black Forest", recently seen in the news?
a) An initiative aimed at reducing black carbon emissions by controlling large-scale forest fires.
b) A programme focused on improving livelihood opportunities for Denotified Tribes.
c) A coordinated drive to remove invasive plant species from Indian forest ecosystems.
d) A specialised anti-Naxal military operation conducted by Indian security forces.

Ans) d
Exp) Option d is the correct answer.
Operation Black Forest is the largest anti-Naxal operation conducted on the Karregutta hill, located along the Chhattisgarh–Telangana border. It was a joint operation of the Central Reserve Police Force (CRPF), Chhattisgarh Police, District Reserve Guard (DRG), and Commando Battalion for Resolute Action (CoBRA)
The joint forces succeeded in destroying a major Naxal base camp, including their material dumps and supply chains on Karreguttalu hill. The security forces neutralized 27 dreaded Maoists
Source: SFG Essential- Current Affairs- Quarterly Compilation (July–September 2025)- Page no: 45
Subject:) Current Affairs
Topic:) Security
Subtopic:)

Q.21) Consider the following statements:
I. Carbon dioxide (CO_2) emissions in India are less than 0.5 t CO_2/capita.
II. In terms of CO_2 emissions from fuel combustion, India ranks second in the Asia-Pacific region.
III. Electricity and heat producers are the largest sources of CO_2 emissions in India.
Which of the statements given above is/are correct?
a) I and III only
b) II only
c) II and III only
d) I, II and III

Ans) c
Exp) Option c is the correct answer.
Statement I is incorrect: In 2022, India's per capita emissions were 1.89 t CO2 per person. Per capita carbon dioxide (CO2) emissions in India have soared in recent decades, climbing from 0.4 metric tons per person in 1970 to a **high of 2.07 metric tons per person in 2023.**
Statement II is correct: India ranks second in the Asia- Pacific region in terms of CO2 emissions from fuel combustion. China ranks first in the region, and India is second with 2,516.967 Mt CO2 emissions, according to data from the International Energy Agency (IEA).
Statement III is correct: In India, electricity and heat production are the largest source of CO2 emissions. Specifically, the power sector, which is heavily reliant on coal, contributes significantly to the nation's overall CO2 emissions, which directly impact climate change. In 2019, the power sector was responsible for about 70% of India's total CO2 emissions, making it the single largest contributor.
Source:) UPSC CSE Pre 2025

Subject:) Environment
Topic:) Environmental Pollution
Subtopic:)

Q.22) Consider the following statements:
Plastic pollution has a negative health impact because the pollutants in plastic pollution can be:
I. Carcinogenic
II. Neurotoxicant
III. Endocrine disruptor
Which of the statements given above are correct?
a) I and II only
b) I and III only
c) II and III only
d) I, II and III

Ans) d
Exp) Option d is the correct answer
The non-biodegradable character of plastics is a serious challenge. There are health impacts at each stage of the plastics life cycle. Plastics have thousands of hazardous additives (including carcinogens, neurotoxicants, endocrine disruptors) that can leach from products and waste, and persist in the environment, threatening humans and ecosystems. Studies also proved that plastics have affected reproductive abilities in animals.
Some of the compounds associated with plastic life cycle include are: vinyl chloride, dioxins, benzene and bisphenol-A.
Statement I is correct: Plastic pollution poses a significant, growing carcinogenic risk due to toxic additives like BPA and phthalates, and the ingestion of microplastics, which can lead **to chronic inflammation and cancer. Thus, they are considered carcinogenic.**
Statement II is correct: Plastic pollution introduces neurotoxic chemicals, including phthalates and BPA, that threaten both marine and human health. Research indicates that microplastics can cross the blood-brain barrier and cause behavioral changes and oxidative damage.
Statement III is correct: The plastics in our lives are not chemically neutral. They often contain Endocrine Disruptor Chemicals such as:
Bisphenol A (BPA) and Bisphenol-S (BPS): Used in water bottles, food containers, and thermal paper.
Phthalates (e.g., DEHP, DBP): Used to soften plastics and found in cosmetics, toys and IV tubing. - PFAS (Per- and polyfluoroalkyl substances): Found in food packaging and non-stick cookware.
These chemicals mimic or block natural hormones such as estrogen, testosterone, thyroid hormones, and cortisol. They interfere with receptor binding, disrupt gene expression in reproductive organs, and induce oxidative stress, inflammation, and apoptosis (cell death).
Source: https://www.thehindu.com/sci-tech/energy-and-environment/how-serious-is-the-global-plastic-pollution-crisis-explained/article70048859.ece
https://www.thehindu.com/sci-tech/health/endocrine-disruptors-in-plastic-waste-a-new-public-health-threat/article69751185.ece
Subject:) Environment
Topic:) Environmental Pollution
Subtopic:)

Q.23) A contaminated industrial site contains complex pollutants that cannot be effectively degraded by the indigenous microbial population. To enhance the cleanup process, genetically modified bacteria are introduced into the soil to accelerate detoxification.
Which one of the following techniques is best described in the above scenario?
a) Biosparging
b) Bioaugmentation
c) Bioventing

d) Bioslurping

Ans) b

Exp) Option b is the correct answer.

Bioremediation involves the use of **microorganisms to degrade pollutants** in contaminated soil and water. Depending on site conditions and the capability of native microbes, different **in-situ techniques** such as biosparging, bioventing, bioaugmentation, and natural attenuation are used.

Option a is incorrect. Biosparging involves **injecting air or oxygen into the saturated zone (below water table)** to increase oxygen availability and stimulate **native microorganisms** for degradation. It is mainly used for petroleum hydrocarbons like diesel and jet fuels.

Option b is correct. Bioaugmentation refers to the **introduction of specific or genetically modified microorganisms** into a contaminated site to degrade pollutants that indigenous microbes cannot handle. For instance, genetically engineered bacteria like Pseudomonas species are introduced to degrade **oil spills or toxic industrial chemicals**.

Option c is incorrect. Bioventing is an in-situ method where **air is supplied to the unsaturated zone (above water table)** to stimulate aerobic degradation by native microbes. It is commonly used for **volatile organic compounds (VOCs)** and petroleum hydrocarbons. For example, supplying oxygen to soil contaminated with gasoline vapors.

Option d is incorrect. Bioslurping is the technique of **vacuum-enhanced dewatering technolog**y for the remediation of **hydrocarbon-contaminated site**s. It is the combination of two bioremedial processes, **bioventing and vacuum-enhanced free-product recovery from groundwater and soil.** The bioventing process stimulates the aerobic bioremediation of hydrocarbon-contaminated soils in situ. Vacuum-enhanced free-product recovery extracts light non-aqueous phase liquids (LNAPLs) from the capillary fringe and the water table.

Source:) https://egyankosh.ac.in/bitstream/123456789/95596/1/Unit-10.pdf

ForumIAS red book- Chapter-Other Pollutants, page 78

Subject:) Environment

Topic:) Various Environmental Issues

Subtopic:)

Q.24) With reference to pollutants, their sources, and associated health impacts, consider the following information:

Pollutant	**Source**	**Associated Health Disorder**
I. Arsenic	Coal-based Thermal power plants	Knock-knee syndrome
II. Nitrates	Untreated sewage	Blue Baby Syndrome
III. Uranium	Phosphate fertilizers	Itai-Itai disease
IV. Fluoride	Aluminium smelting	Skeletal Fluorosis

In which of the above rows is the given information correctly matched?

a) Only one

b) Only two

c) Only three

d) All the four

Ans) b

Exp) Option b is the correct answer.

Row I is incorrect. Coal mining and coal combustion in thermal power plants (TPPs) are the major anthropogenic sources of arsenic (As) contamination in many different industrial regions. Arsenic exposure causes skin lesions, cancers, and arsenicosis, **not knock-knees syndrome**. Knock-knees are typically linked with fluoride toxicity.

Row II is correct. Nitrates commonly enter water bodies through **untreated sewage, agricultural runoff, and fertilizers**. High nitrate levels in drinking water can cause **Blue Baby Syndrome** (methemoglobinemia) in infants, where oxygen transport in blood is impaired, leading to a bluish discoloration of the skin.

Row III is incorrect. Trace amounts of **uranium may be present in phosphate fertilizer**s but **Itai-Itai disease is caused by cadmium poisoning,** not uranium. Uranium is mainly associated with kidney damage and carcinogenic risks.

Row IV is correct. Fluoride can be released from industrial activities such as **aluminium smelting,** and excessive fluoride intake leads to **skeletal fluorosis,** characterized by joint stiffness, bone deformities, and in severe cases, **knock-knees.**

Source:) https://egyankosh.ac.in/bitstream/123456789/86869/1/Unit-12.pdf 246, 252, 254
https://egyankosh.ac.in/bitstream/123456789/13744/1/Unit-11.pdf page 26
https://egyankosh.ac.in/bitstream/123456789/91797/1/Unit-6.pdf under heading nitrogen
ForumIAS red book-chapter-Air & Water Pollution, page 45
https://indianexpress.com/article/cities/delhi/heavy-metal-contamination-in-delhi-flagged-in-cpcb-report-uranium-levels-3rd-highest-in-country-10392309/
https://www.newindianexpress.com/nation/2018/Jun/08/scientists-say-there-is-severe-uranium-contamination-in-indias-groundwater-1825487.html
Subject:) Environment
Topic:) Environmental Pollution
Subtopic:)

Q.25) Which one of the following air pollutants combines with the haemoglobin of human blood and reduces its oxygen-carrying capacity, leading to suffocation and may cause even death?
a) Chlorofluorocarbon
b) Fly ash
c) Carbon monoxide
d) Sulphur dioxide

Ans) c
Exp) Option c is the correct answer.
The air pollutant that combines with the haemoglobin of human blood and reduces its oxygen carrying capacity, leading to suffocation and potential death is Carbon monoxide. Carbon monoxide can c**ombine with the haemoglobin of human blood and form carboxyhaemoglobin,** which reduces the oxygen-carrying capacity of the blood. This can lead to hypoxia, which is a condition where the tissues and organs do not receive enough oxygen.
Source: UPSC CDS [II] 2016
Subject:) Environment
Topic:) Environmental Pollution
Subtopic:)

Q.26) In the context of global environmental pollution, the term "Grasshopper Effect" refers to:
a) Rapid biodegradation of pollutants in tropical regions.
b) Accumulation of pollutants in soil due to agricultural practices.
c) Long-range transport of persistent pollutants through repeated evaporation and condensation cycles.
d) The sudden increase in local pollutant concentration due to the migratory patterns of insects.

Ans) c
Exp) Option c is the correct answer.
Also known as global distillation, the Grasshopper Effect is a geochemical process where **certain pollutants** (specifically, Persistent Organic Pollutants (POPs) like DDT and PCBs) **are transported over long distances from warmer tropical and temperate regions to colder polar regions.** These chemicals are **semi-volatile**, meaning **they evaporate into the atmosphere in warm climates**. They are then carried by air currents toward the poles. **When they encounter colder temperatures, they condense** and settle back to the Earth's surface through rain, snow, or dust.
Source:) https://www.downtoearth.org.in/waste/run-up-to-ottawa-arctic-a-hemispheric-sink-for-chemicals-plastics-13-million-people-in-region-at-risk-says-report-95604
https://www.unep.org/news-and-stories/story/grasshopper-effect-serves-pollutants-plates-arctic-peoples

Subject:) Environment
Topic:) Environmental Pollution
Subtopic:)

Q.27) With reference to fly ash, consider the following statements:
I. Its composition includes silicon dioxide (SiO_2), calcium oxide (CaO), and aluminum oxide (Al_2O_3).
II. ASHTRACK is a web portal and app for better management of fly ash produced by thermal power plants.
III. It can be utilized as a soil conditioner to improve the water-holding capacity of soils.
IV. Fly ash bricks manufacturing units in India are listed under the red category of Industries as per the Central Pollution Control Board classification of industries based on Pollution index.
Which of the statements given above are correct?
a) I and II only
b) III and IV only
c) I, II and III only
d) I, II, III and IV

Ans) c
Exp) Option c is the correct answer.
Statement I is correct: Fly ash is a fine, powdery residue generated as a by-product of coal combustion in thermal power plants. The term "fly ash" includes all ash generated such as Electrostatic Precipitator (ESP) ash, dry fly ash, bottom ash, pond ash and mound ash. Its **composition, rich in silicon dioxide (SiO2), calcium oxide (CaO), and aluminum oxide (Al2O3)**.
Statement II is correct: ASHTRACK is a mobile application and web portal launched by the Ministry of Power for better management of fly ash produced by thermal power plants by providing an interface between fly ash producers (Thermal Power Plants) and potential ash users such as – road contractors, cement plants etc.
Statement III is correct: In agriculture, fly ash is effectively utilized as a **soil conditioner**. It helps **improve the water-holding capacity** of the soil and enhances its aeration and texture. Additionally, it provides essential micronutrients like iron, magnesium, and potassium to the crops.
Statement IV is incorrect: As per the Central Pollution Control Board classification of industries based on the pollution index, fly ash brick manufacturing units without boilers in India are listed under green and manufacturing units with boilers are classified under orange category.
Source:) https://www.thehindu.com/news/cities/Visakhapatnam/extracting-rare-earth-elements-from-fly-ash-is-akin-to-searching-needle-in-a-hay-stack-say-researchers/article70556435.ece
https://www.thehindu.com/news/cities/Visakhapatnam/call-for-focus-on-fly-ash-brick-industry/article68539437.ece
https://www.pib.gov.in/PressReleasePage.aspx?PRID=2160714&reg=3&lang=2
https://cpcb.nic.in/openpdffile.php?id=TGF0ZXN0RmlsZS9fMTczNzYxMzk2OV9tZWRpYXBob3RvMTEzODMucGRm (Page 32)
Subject:) Environment
Topic:) Various Environmental Issues
Subtopic:)

Q.28) With reference to Chromium, consider the following statements:
I. The leather tanning industry is a major source of chromium pollution.
II. Chromium contamination in agricultural soil reduces soil microbial activity.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

Ans) c

Exp) Option c is the correct answer.

Statement I is correct: Chromium compounds, especially hexavalent chromium, are extensively used in leather tanning processes. **Effluents from tanning industries are a major contributor to chromium contamination in water and soil. This leads to severe groundwater and soil contamination.**

Statement II is correct: Chromium contamination adversely affects soil microorganisms by disrupting enzymatic activity and metabolic functions. This leads to reduced microbial population and activity, impairing soil fertility and nutrient cycling. High concentrations of chromium inhibit the growth of essential soil bacteria and fungi.

Source:) https://www.downtoearth.org.in/waste/using-tannery-waste-in-fields-adding-toxics-to-soil-study-57912
https://www.thehindu.com/news/national/uttar-pradesh/adequate-safe-drinking-water-not-available-in-chromium-contaminated-areas-of-up-ngt/article69461886.ece
https://cdn.who.int/media/docs/default-source/wash-documents/wash-chemicals/chromium.pdf?sfvrsn=37abd598_6 Pg:1
https://www.thehindu.com/sci-tech/chromium-pollution
Subject:) Environment
Topic:) Environmental Pollution
Subtopic:)

Q.29) Under the E-Waste (Management) Amendment Rules, 2024, who among the following shall fix the highest and lowest price for exchange of extended producer responsibility (EPR) certificates on the platform for exchange or transfer of EPR certificates?

a) Bureau of Indian Standards (BIS)
b) Securities and Exchange Board of India (SEBI)
c) Central Pollution Control Board (CPCB)
d) The Energy and Resources Institute (TERI)

Ans) c

Exp) Option c is the correct answer.:

As per the E-Waste (Management) Amendment Rules, 2024, the Central Government may, by order, establish one or more platforms for the exchange or transfer of Extended Producer Responsibility (EPR) certificates. The operation of these platforms is regulated in accordance with the recommendations of the Central Pollution Control Board (CPCB). The CPCB is responsible for the technical framework, including the generation, verification, and audit of these certificates.

The Central Pollution Control Board shall fix the highest (100%) and lowest price (30%) for exchange of extended producer responsibility certificates of the environmental compensation for non-fulfilment of the extended producer responsibility obligation under the E-Waste (Management) Amendment Rules, 2024

Source:) https://cpcb.nic.in/uploads/Projects/E-Waste/E-waste-Third-Amendment-Rules-2024.pdf (page 2-3)
Subject:) Environment
Topic:) Environmental Pollution
Subtopic:)

Q.30) Consider the following statements regarding the World Food Programme (WFP):

I. Food and Agriculture Organization was instrumental in the establishment of the World Food Programme.
II It is one of the agencies of the United Nations.
III. Its operations are entirely funded through the voluntary contributions from donors.
IV. It was awarded the Nobel Peace Prize in 2020.

Which of the statements given above are correct?

a) I and III only
b) I, II and III only
c) II and IV only
d) I, II, III and IV

Ans) d

Exp) Option d is the correct answer.

India recently launched a new collaborative initiative with the support of the UN World Food Programme to strengthen Rice Fortification and Supply Chain Management in Nepal. This initiative is part of India-UN Global Capacity Building Initiative.

Statement I is correct. WFP was established in 1961 by the United Nations General Assembly & Food and Agriculture Organization (FAO).

Statement II is correct. The World Food Programme (WFP) is a UN agency dedicated to ending hunger, achieving food security, and improving nutrition.

World Food Programme (WFP) is an international organization within the United Nations (UN) that provides food assistance worldwide. The World Food Programme aims to eradicate hunger and malnutrition.

It is an executive member of the United Nations Sustainable Development Group, with a priority to achieve SDG 2, "zero hunger", by 2030. It is headquartered in Rome and has offices in 87 countries.

Statement III is correct. Its operations are **entirely funded through the voluntary contributions** of donor governments, institutions, corporations and individuals.

Statement IV is correct. The World Food Programme (WFP) was awarded the Nobel Peace Prize in 2020.

Source: SFG Essential, Current Affairs, Quarterly Compilation (July–September 2025), Page: 29, 30

Subject:) Current Affairs

Topic:) International Relations

Subtopic:)

Q.31) Which of the following is/are the possible consequence/s of heavy sand mining in river beds?

1. Decreased salinity in the river
2. Pollution of groundwater
3. Lowering of the water table

Select the correct answer using the code given below:

a) 1 only

b) 2 and 3 only

c) 1 and 3 only

d) 1, 2 and 3

Ans) b

Exp) Option b is the correct answer.

Option 1 is incorrect: Decreased salinity is not a likely consequence of heavy sand mining in riverbeds. In fact, heavy sand mining can actually **increase the salinity** of rivers by removing sand, which acts as a natural filter.

Option 2 is correct: Pollution of groundwater is a very likely consequence of heavy sand mining in riverbeds. Sand mining can disturb and pollute groundwater aquifers, especially in areas with shallow aquifers. This can lead to contamination of drinking water supplies with heavy metals, chemicals, and other pollutants.

Option 3 is correct: Lowering of water table is also a very likely consequence of heavy sand mining in riverbeds. Sand mining can lower the water table by removing sand, which acts as a sponge that stores water. This can lead to water shortages and other problems for communities that rely on groundwater for drinking and irrigation.

Source: UPSC CSE Pre 2018

Subject:) Environment

Topic:) Various Environmental Issues

Subtopic:)

Q.32) Consider the following statements with respect to the C-FLOOD:

I. It is a web-based platform that provides two days' advance inundation forecasts up to the village level.

II. It provides information about storage levels of water in important reservoirs in the country.

Which of the statements given above is/are correct?

a) I only

b) II only

c) Both I and II
d) Neither I nor II

Ans) a
Exp) Option a is the correct answer.
Statement I is correct: C-FLOOD (Coastal Flood Warning System) a web-based platform. It provides two days' advance inundation forecasts up to village level in the form of inundation maps and water level predictions. Some of its features are:

- C-Flood integrates flood inundation output information derived from advanced 2D hydrodynamic modelling with extent and depth information.
- It has been designed to serve as a unified inundation information system integrating flood modelling outputs from national and regional agencies for all river basins as per their respective action plans. It covers inundation forecasts for the Godavari, Tapi and Mahanadi River Basins in the initial phase.

Statement II is incorrect: Central Water Commission (CWC) monitors the live storage position of 161 important reservoirs spread all over the country and issues bulletins regarding the same every Thursday.
Source:) https://www.pib.gov.in/PressReleasePage.aspx?PRID=2159056&reg=3&lang=2
https://www.pib.gov.in/PressReleasePage.aspx?PRID=2141608&reg=3&lang=2#:~:text=Patil%20inaugurated%20C%2DFLOOD%2C%20a,of%20Jal%20Shakti%20on%20today.
https://www.jalshakti-dowr.gov.in/static/uploads/2025/08/126ea6a5fc404eaba643f48e281c47ff.pdf
Subject:) Environment
Topic:) Disaster Management
Subtopic:)

Q.33) In the context of global risk management, which of the following best describes a 'Grey Rhino' event?
a) A highly improbable event that has massive impacts and is rationalized in hindsight.
b) A highly probable, high-impact threat which was ignored despite visible warning signs.
c) A gradual and insignificant risk that has minimal long-term impact.
d) A sudden, localized surge in pollution levels in an area due to industrial accidents.

Ans) b
Exp) Option b is the correct answer.
A 'Grey Rhino' event refers to a highly probable and high-impact threat that is obvious and visible, yet neglected or insufficiently addressed by decision-makers. Unlike unexpected "Black Swan" events, Grey Rhino risks are known dangers. Ex: climate change.
It contrasts with "black swan events" (rare, unpredictable) and focuses on obvious but ignored risks. This framework emphasizes that disasters can be averted through proactive risk management, strict environmental regulations, and acting on ecological reports before they become catastrophes.
Source:) https://www.thehindu.com/news/national/kerala/wayanad-landslides-a-highly-probable-grey-rhino-event-ignored-despite-visible-warning-signs-says-study/article70041440.ece
https://timesofindia.indiatimes.com/blogs/toi-edit-page/we-are-all-sitting-ducks-now/
https://www.downtoearth.org.in/natural-disasters/grey-rhino-in-the-western-ghats-kerala-ignored-warnings-before-wayanads-deadly-landslide-says-new-report
Subject:) Environment
Topic:) Various Environmental Issues
Subtopic:)

Q.34) Consider the following pollutants:
I. Sulphur dioxide (SO_2)
II. Nitrogen oxides (NO_x)
III. Methane (CH_4)
IV. Ammonia (NH_3)

Which of the pollutants given above are included under the Gothenburg Protocol which aims at reducing long-range transboundary air pollution?
a) I and II only
b) III and IV only
c) I, II and IV only
d) I, II, III and IV

Ans) c
Exp) Option c is the correct answer.
The **Gothenburg Protocol** is a multi-pollutant protocol **adopted in 1999 under the framework of the Convention on Long-Range Transboundary Air Pollution** (CLRTAP), overseen by the United Nations Economic Commission for Europe (UNECE).
It aims to control and reduce emissions of air pollutants that cause:

- **Acidification** of soils and water bodies
- **Eutrophication** of ecosystems (due to excess nitrogen)
- **Formation of ground-level ozone** (smog)

Pollutants Targeted Under the Protocol:

1. **Sulphur Dioxide** (SO_2) – major cause of acid rain.
2. **Nitrogen Oxides** (NO_x) – contributes to both acid rain and smog.
3. **Ammonia** (NH_3) – contributes to eutrophication and formation of fine particulate matter (PM2.5).
4. **Volatile Organic Compounds** (VOCs) – precursors of ground-level ozone.

Methane (CH_4) is not covered under the Gothenburg Protocol.
Source: https://unece.org/environmental-policy/air/protocol-abate-acidification-eutrophication-and-ground-level-ozone
https://www.canada.ca/en/environment-climate-change/corporate/international-affairs/partnerships-organizations/gothenburg-protocol-air-pollution.html
Subject:) Environment
Topic:) Environmental Pollution
Subtopic:)

Q.35) With reference to Chemical Oxygen Demand (COD), consider the following statements:
I. It is a measure of the oxygen equivalent required to chemically oxidize total organic matter present in water.
II. COD values are generally lower than Biological Oxygen Demand (BOD) values for the same water sample.
III. In distilled water, the COD value is zero or nearly zero.
Which of the statements given above is/are correct?
a) I and III only
b) I only
c) II and III only
d) III only

Ans) a
Exp) Option a is the correct answer.
Statement I is correct. Chemical Oxygen Demand (COD) is an important indicator of **water quality and pollution level**. It measures the **total oxygen required to chemically oxidize all oxidizable substances** in water, including both **biodegradable and non-biodegradable organic matter.**
Statement II is incorrect. COD values are **always higher than BOD values** for a given sample because COD includes oxidation of **both biodegradable and non-biodegradable substances**, whereas BOD measures only biodegradable organic matter.
Statement III is correct. Distilled water is free from organic pollutants and most inorganic reducing substances. Since COD depends on the presence of oxidizable material, **pure distilled water shows negligible or zero COD**. This is why distilled water is often used as a **blank sample in COD experiments**.

Source:) https://egyankosh.ac.in/bitstream/123456789/31144/1/Exp-11.pdf page 35,36
ForumIAS red book-chapter-Air & Water Pollution, page 45
Subject:) Environment
Topic:) Various Environmental Issues
Subtopic:)

Q.36) The 'Global Drought Outlook, 2025' report was released by which of the following organizations?
a) United Nations Convention to Combat Desertification (UNCCD)
b) World Meteorological Organization (WMO)
c) Organisation for Economic Cooperation and Development (OECD)
d) Food and Agriculture Organization (FAO)

Ans) c
Exp) Option c is the correct answer.
The 'Global Drought Outlook, 2025' is released by the Organisation for Economic Co-operation and Development (OECD). It provides a comprehensive, evidence-based assessment of how drought trends and impacts are evolving under climate change and identifies policy responses for effective adaptation.
According to the report, countries like India, Australia, etc. may face water-related disruptions in the operations of hydroelectric power stations. Crop yields can decline by up to 22% in India. There will be 3% - 7.5% annual increase in the economic cost of an average drought event in countries like India.
Source:) https://www.oecd.org/en/publications/global-drought-outlook_d492583a-en/full-report.html
Subject:) Environment
Topic:) Various Environmental Issues
Subtopic:)

Q.37) With reference to Methane, consider the following statements:
I. It is a short-lived climate pollutant.
II. It is the second-largest contributor to global warming after carbon dioxide.
III. The Global Methane Pledge targets a minimum 60% reduction in global methane emissions from 2020 levels by 2030.
Which of the statements given above are correct?
a) I and II only
b) II and III only
c) I and III only
d) I, II and III

Ans) a
Exp) Option a is the correct answer.
Methane is a potent greenhouse gas that plays a critical role in the Earth's climate system.
Statement I is correct: Methane is much more effective at trapping heat than carbon dioxide; its **Global Warming Potential (GWP) is over 80 times that of CO2 over a 20-year period**. However, it has a much **shorter atmospheric lifespan (approximately 12 years)** compared to carbon dioxide, which can persist for centuries.
Statement II is correct: Methane is the second-largest contributor to global warming after carbon dioxide. Methane is the second most abundant anthropogenic greenhouse gas after carbon dioxide (CO2). It is responsible for approximately **30% (nearly one-third)** of the rise in global temperatures since the Industrial Revolution.
Statement III is incorrect: Launched at COP26, the Global Methane Pledge is a voluntary international initiative aimed at cutting global methane emissions by at **least 30% from 2020 levels by 2030**. India, the world's third-largest methane emitter.
Source:) What is India's stance on methane emissions, potent greenhouse gas?
https://www.thehindu.com/news/cities/mumbai/un-report-flags-indias-rising-methane-emissions-highlights-agricultural-gaps/article70298356.ece
Subject:) Environment

Topic:) Environmental Pollution
Subtopic:)

Q.38) Consider the following substances:
I. Chlorofluorocarbons (CFCs).
II. Hydrochlorofluorocarbons (HCFCs).
III. Hydrofluorocarbons (HFCs).
IV. Carbon tetrachloride.
How many of the above are major Ozone Depleting Substances (ODS)?
a) Only one
b) Only two
c) Only three
d) All the four

Ans) c
Exp) Option c is the correct answer.
Option I is correct: Chlorofluorocarbons are **highly stable chemicals** that do not break down in the lower atmosphere. Once they reach the stratosphere, ultraviolet radiation dissociates them, releasing chlorine atoms that destroy ozone molecules. **They are characterized by a high Ozone Depletion Potential (ODP).**
Option II is correct: Hydrochlorofluorocarbons contain hydrogen, which **makes them less stable**. A significant portion breaks down in the lower atmosphere before reaching the ozone layer. Hence, they **have a much lower ODP (0.01 to 0.1).**
Option III is incorrect: Hydrofluorocarbons contain no chlorine or bromine atoms, which are the primary catalysts for ozone destruction. As they do not participate in the chemical reactions that deplete the ozone layer, **their Ozone Depletion Potential is near zero.**
Option IV is correct: Carbon tetrachloride is a powerful ozone-depleting substance that was widely used as a solvent. Like CFCs, it **is very stable and releases large amounts of chlorine** into the stratosphere. It has **a high Ozone Depletion Potential (ODP).**
Source:) https://egyankosh.ac.in/bitstream/123456789/86845/1/Unit-3.pdf#:~:text=Some%20chemical%20compounds%20like%20Methane,are%20dissociated%20by%20the%20stratospheric (page 42-44)
https://www.egyankosh.ac.in/bitstream/123456789/116069/1/Unit-11.pdf#:~:text=These%20HFCs%20were,not%20deplete%20the%20ozone%20layer%2C (page 253)
Subject:) Environment
Topic:) Environmental Pollution
Subtopic:)

Q.39) The term 'LICOMK++', which was recently in the news, is best described as which of the following?
a) A high-performance supercomputing cluster for weather forecasting in the Himalayan region.
b) A high-resolution global ocean simulation system used for studying ocean dynamics.
c) An AI-based platform for predicting monsoon variability in tropical regions.
d) A deep-sea exploration vehicle designed for mapping ocean floors.

Ans) b
Exp) Option b is the correct answer.
Chinese researchers have recently developed a high-resolution ocean simulation system called LICOMK++ to better study ocean and climate patterns. **LICOMK++ is a high-resolution global ocean general circulation model (OGCM). It is a high-resolution global ocean circulation model used for studying ocean dynamics, heat transport, and climate interactions.** It is primarily used in climate research and numerical simulations of ocean processes.

Source:) https://www.business-standard.com/world-news/china-unveils-advanced-ocean-simulator-licomk-amid-us-chip-sanctions-125051200565_1.html https://indianexpress.com/article/explained/licomk-plus-plus-ocean-model
Subject:) Environment
Topic:) Various Environmental Issues
Subtopic:)

Q.40) Consider the following pairs:

Area of conflict mentioned in News	Country where it is located
I. Deir el-Balah	Sudan
II. Khartoum	Somalia
III. Donetsk	Ukraine

How many of the pairs given above are correctly matched?
a) Only one
b) Only two
c) All the three
d) None

Ans) a
Exp) Option a is the correct answer.
Pair I is incorrect. Deir al-Balah is a city of Gaza, Palestine. The United Nations has declared a famine in Gaza in August 2025, as nearly 500,000 people are at risk of starvation.
Pair II is incorrect. Khartoum is in Sudan. Sudan descended into civil war in April 2023 due to a violent power struggle between the national army and the paramilitary Rapid Support Force (RSF).
Pair III is correct. Donetsk is an industrial city in Ukraine. Parts of Donetsk are now controlled by Russia.
Source: SFG Essential, Current Affairs, Quarterly Compilation (July–September 2025), Page: 34
Subject:) Current Affairs
Topic:) International Relations
Subtopic:)

Q.41) With reference to 'Direct Air Capture', an emerging technology, which of the following statements is/are correct?
I. It can be used as a way of carbon sequestration.
II. It can be a valuable approach for plastic production and in food processing.
III. In aviation, it can be a source of carbon for combining with hydrogen to create synthetic low-carbon fuel.
Select the correct answer using the code given below.
a) I and II only
b) III only
c) I, II and III
d) None of the above statements is correct

Ans) c
Exp) Option c is the correct answer.
Statement I is correct: Direct air capture (DAC) is a method that can be used for carbon sequestration, also known as direct air carbon capture and sequestration (DACCS). DACCS involves extracting carbon dioxide (CO_2) directly from the atmosphere using chemical or physical processes, followed by the long-term storage of that captured CO_2, effectively removing it from the atmosphere.

Statement II is correct: Direct air capture (DAC) can be a valuable approach for certain aspects of plastic production and food processing by utilizing captured CO2. DAC extracts CO2 from the atmosphere, which can then be used as a feedstock for producing some types of plastics and in food production processes.
Statement III is correct: Direct air capture (DAC) technology can be used in aviation to create synthetic low-carbon fuels. DAC extracts carbon dioxide (CO2) from the atmosphere. This captured CO2 can then be combined with hydrogen, which can be produced through renewable energy sources, to create synthetic fuels like e-fuels
Source: UPSC CSE Pre 2025
Subject:) Environment
Topic:) Various Environmental Issues
Subtopic:)

Q.42) The "Global Alliance to Eliminate Lead Paint" is established by:
a) United Nations Environment Programme (UNEP) and the World Health Organization (WHO)
b) United Nations Development Programme (UNDP) and World Bank (WB)
c) Food and Agriculture Organization (FAO) and United Nations Environment Programme (UNEP)
d) World Bank (WB) and United Nations Environment Programme (UNEP)

Ans) a
Exp) Option a is the correct answer.
The Global Alliance to Eliminate Lead Paint (Lead Paint Alliance) is a global voluntary partnership established to prevent human exposure to lead from paints. This partnership was formed **jointly by the United Nations Environment Programme (UNEP) and the World Health Organization (WHO).** The primary objective of the alliance is to promote the phase-out of the manufacture and sale of paints containing lead and to eliminate the health risks associated with lead exposure, particularly for children and workers.
Source:) https://www.unep.org/topics/chemicals-and-pollution-action/chemicals-management/pollution-and-health/heavy-metals/global
https://www.thehindu.com/news/national/other-states/This-festive-season-make-your-home-lead-free/article16071760.ece
Subject:) Environment
Topic:) Various Environmental Issues
Subtopic:)

Q.43) With reference to Photochemical smog, consider the following statements:
Statement I: It is generally more prevalent during the summer months compared to winter.
Statement II: It is oxidising in nature.
Statement III: The increased solar radiation and high temperatures during the summer months drive the necessary chemical reaction for its formation.
Which one of the following is correct in respect of the above statements?
a) Both Statement II and Statement III are correct and both of them explain Statement I
b) Both Statement II and Statement III are correct but only one of them explains Statement I
c) Only one of the Statements II and III is correct and that explains Statement I
d) Neither Statement II nor Statement III is correct

Ans) b
Exp) Option b is the correct answer.
Statement I is correct: Photochemical smog occurs in warm, dry, and sunny climates. Thus the photochemical smog is more prevalent during the summer months compared to winter.
Statement II is correct: Photochemical smog is characterized by a **high concentration of oxidizing agents, primarily ozone (O3), peroxyacetyl nitrate (PAN), and nitric acid**. Because these components are capable of oxidizing other substances, the smog is fundamentally oxidizing in nature.
Statement II does not explain Statement I: Being oxidizing or reducing does not explain why the photochemical smogs are prevalent during the summer.

Statement III is correct: Photochemical smog formation is highly dependent on two key factors: intense sunlight and high temperatures. The summer months provide both conditions optimally. **The increased solar radiation drives the necessary chemical reactions**, such as the photolysis of nitrogen dioxide, more vigorously. Higher temperatures also accelerate the reaction rates, leading to faster and greater accumulation of ground-level ozone and other secondary pollutants.

Statement III explains Statement I: Photochemical Smog is generally more prevalent during the summer months compared to winter. This is because **intense solar radiation is a necessary catalyst** for the chemical reactions between primary pollutants, such as nitrogen oxides (NOx) and volatile organic compounds (VOCs).

Source:)
https://egyankosh.ac.in/bitstream/123456789/64015/1/Unit-9.pdf (page 99)
https://egyankosh.ac.in/bitstream/123456789/13541/1/Unit-9.pdf#:~:text=Photochemical%20smog,The%20primary%20pollutants%20involved%20are (page 52-53)
Subject:) Environment
Topic:) Environmental Pollution
Subtopic:)

Q.44) With reference to the possible effects of Acid Rain, consider the following statements:
I. The leaching of essential plant nutrients such as calcium and magnesium from the soil.
II. The destruction of beneficial nitrogen-fixing bacteria in the soil, thereby hindering plant growth.
III. Enhancing dissolved organic carbon (DOC) levels in lakes, thereby reducing UV penetration.
IV. Deterioration of exposed limestone and marble-based structures.
Which of the statements given above are correct?
a) I, II and III only
b) I and II only
c) I, II and IV only
d) I, II, III and IV

Ans) c
Exp) Option c is the correct answer.
Statement I is correct: Acid rain increases the acidity of the soil, which triggers a process where essential nutrients such as **calcium**, **magnesium**, and potassium a**re leached (washed away) from the topsoil.**
Statement II is correct: High acidity in the soil can **kill or inhibit beneficial microorganisms,** including nitrogen-fixing bacteria.
Statement III is incorrect: Acid rain actually **decreases the concentration of dissolved organic carbon (DOC) in lakes**. Because DOC acts as a natural "sunscreen" that blocks harmful radiation, its reduction leads to **increased UV penetration deeper into the water.**
Statement IV is correct: Acid rain is **highly damaging to marble and limestone structures.** It reacts with the calcium carbonate in the stone to form gypsum, which is soluble and flakes off easily. This causes corrosion and discoloration of monuments.
Source:) https://www.egyankosh.ac.in/bitstream/123456789/79938/1/Block-3.pdf (Pg: 251)
https://www.thehindu.com/news/cities/Delhi/pollution-impact-on-monuments-slow-but-long-lasting-experts/article8487562.ece
Subject:) Environment
Topic:) Environmental Pollution
Subtopic:)

Q.45) Which one of the following is the best description of "INS Udayagiri" and "INS Himgiri" which were seen recently in the news?
a) Nuclear-powered submarine
b) Torpedo launch and recovery vehicle
c) Coastal surveillance vessels
d) Stealth guided-missile frigates

Ans) d

Exp) Option d is the correct answer.

INS Udayagiri and INS Himgiri are the latest Project 17A (P-17A) multi-mission stealth frigates commissioned by the Indian Navy at Naval Base Visakhapatnam. The P-17A class represents a major technological upgrade over the earlier P17 Shivalik-class, featuring enhanced stealth shaping, an improved hull design, and upgraded weapon and sensor suites.

A total of seven frigates are being constructed under Project 17A such as INS Nilgiri, Udayagiri, Himgiri, Taragiri, Mahendragiri, Dunagiri, and Vindhyagiri, all of which will join the Eastern Fleet, strengthening India's maritime power projection across the Indian Ocean Region (IOR).

These frigates use a modern CODOG propulsion system (Combined Diesel or Gas), enabling high-speed manoeuvrability and long-range endurance. They are equipped with an Integrated Platform Management System, advanced indigenous weapons and sensors, and are capable of performing the full spectrum of maritime operations in blue-water conditions, including anti-air, anti-surface, and anti-submarine warfare.

With around 75% indigenous content, supported extensively by Indian MSMEs, Project 17A is a major milestone in advancing Aatmanirbhar Bharat in defence manufacturing.

Source: SFG Essential- Current Affairs- Quarterly Compilation (July–September 2025- Page no: 47

Subject:) Current Affairs

Topic:) Science & Technology

Subtopic:)

Q.46) How many times do the hour hand and the minute hand form a right angle in a clock between 1:00 p.m. and 5:00 p.m. (same day)?

a) 6 times

b) 7 times

c) 8 times

d) 9 times

Ans) b

Exp) Option b is the correct answer.

For the hands of a clock to form a **right angle**, the angle between them must be: **90°** or **270°**

Let the time be **h : m**, where **h** is the hour and **m** is the minutes.

Then:

- Angle made by the **hour hand** = **30h + 0.5m**
- Angle made by the **minute hand** = **6m**

So the angle difference is: **|30h − 5.5m|**

For a right angle: **|30h − 5.5m| = 90**

This gives: **30h − 5.5m = 90** or **30h − 5.5m = −90**

Now check the interval from **1:00 p.m. to 5:00 p.m.**

Between 1 and 2: Two right angles occur.

Between 2 and 3: One right angle occurs before 3:00, and one occurs exactly at **3:00**.

Between 3 and 4: One new right angle occurs after 3:00.

Between 4 and 5: Two right angles occur.

So total number of times = **2 + 1 + 1 + 1 + 2 = 7**

Subject:) CSAT

Topic:) Quantitative Aptitude

Subtopic:)

Q.47) The total cost of 5 proteins, 8 creatines and 10 aminos is equal to three times the total cost of 1 protein, 3 creatines and 2 aminos. In this context consider the following statements:

I. The total cost of 5 proteins, 4 creatines and 12 aminos is equal to the total cost of 3 proteins, 5 creatines and 8 aminos.

II. The total cost of 1 creatine and 4 aminos is equal to the cost of 2 proteins.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

Ans) a
Exp) Option a is the correct answer.
Let the cost of:

- **1 protein = p**
- **1 creatine = c**
- **1 amino = a**

Given: **5p + 8c + 10a = 3(p + 3c + 2a)**
5p + 8c + 10a = 3p + 9c + 6a
5p − 3p + 8c − 9c + 10a − 6a = 0
2p − c + 4a = 0
So the relation is: **2p − c + 4a = 0** or **2p + 4a = c**
Statement I is correct: 5p + 4c + 12a = 3p + 5c + 8a
Take the difference:
(5p + 4c + 12a) − (3p + 5c + 8a)
= 2p − c + 4a
But from the given relation: **2p − c + 4a = 0**
So both sides are equal.
Statement II is incorrect: c + 4a = 2p
But from the relation obtained: **2p − c + 4a = 0**
This gives: **2p + 4a = c**
Or **c = 2p + 4a**
This is **not the same as**: **c + 4a = 2p**
Subject:) CSAT
Topic:) Quantitative Aptitude
Subtopic:)

Q.48) $125^4 + 25^7$ is divisible by
a) 11
b) 17
c) 65
d) 19

Ans) c
Exp) Option c is the correct answer.
First express both terms as powers of **5**:
$125^4 = (5^3)^4 = 5^{12}$
$25^7 = (5^2)^7 = 5^{14}$
So,
$125^4 + 25^7 = 5^{12} + 5^{14}$
Take 5^{12} common:
$125^4 + 25^7 = 5^{12}(1 + 5^2)$
$= 5^{12}(1 + 25)$
$= 5^{12} \times 26$
$= 5^{12} \times 2 \times 13$

So the expression is divisible by:
5 × 13 = 65
Subject:) CSAT
Topic:) Quantitative Aptitude
Subtopic:)

Directions for the following 2 (two) items:
Read the following passage and answer the items that follow the passage. Your answers to these items should be based on the passage only

Passage - 1

India's rural economy continues to be shaped by the uncertainties of water availability. While large irrigation projects often dominate public discourse, nearly 52 per cent of the country's gross cropped area remains rainfed. In this context, watershed development deserves renewed attention as a central pillar of India's strategy for sustainable rural growth. Structures such as check dams, farm ponds, contour bunds and percolation tanks have been associated with rises in groundwater levels, expansion of irrigated area and increases in cropping intensity often ranging from 10-20 percentage points. Such improvements allow farmers to diversify into horticulture, pulses and other higher value crops. The economic impact of watershed development extends beyond crop output. Improved water availability often stimulates investments in livestock, agro processing and small rural enterprises. Watershed programmes also interact closely with rural infrastructure development. Increased water security encourages the creation of farm level assets, minor irrigation channels and storage facilities. Taken together, these outcomes demonstrate that managing water at the micro catchment level can generate macro economic dividends.

Q.49) Which one of the following statements best reflects the most logical and rational inference that can be drawn from the passage?
a) Watershed development contributes to increased water availability by groundwater recharge mechanisms.
b) Local level interventions are needed to resolve the problem of water scarcity in rural India.
c) Efficient micro-level water management can generate tangible and sustained economic gains for the rural economy.
d) Watershed development reduces the dependence of agricultural output on uncertain rainfall.

Ans) c
Exp) Option c is the correct answer.
Statement a is incorrect. The passage states "Structures such as check dams, farm ponds, contour bunds and percolation tanks have been associated with rises in groundwater levels". However, statement a does not best reflect the most logical and rational inference that can be drawn from the passage as it **focuses on only one part of the passage.**
Statement b is incorrect. While the passage talks about watershed development as a measure to resolve the problem of water scarcity in rural India, statement a does not best reflect the most logical and rational inference that can be drawn from the passage as **it does not talk about watershed development.**
Statement c is correct. It best reflects the most logical and rational inference that can be drawn from the passage as it highlights the role of watershed development in resolving the problem of water scarcity in rural India It is mentioned in the passage as "**managing water at the micro catchment level can generate macro economic dividends**."
Statement d is incorrect. The passage states "While large irrigation projects often dominate public discourse, nearly 52 per cent of the country's gross cropped area remains rainfed. In this context, watershed development deserves renewed attention". However, statement a does not best reflect the most logical and rational inference that can be drawn from the passage as it **focuses on only one part of the passage.**
Subject:) CSAT
Topic:) Reading Comprehension
Subtopic:)

Q.50) With reference to the above passage, the following assumptions have been made:
I. Rural India is experiencing severe water scarcity due to declining groundwater levels.
II. Large scale irrigation projects cannot solve the problem of water scarcity in rural India.
Which of the above assumptions is/are valid?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

Ans) d
Exp) Option d is the correct answer.
Statement I is incorrect. While the passage states "India's rural economy continues to be shaped by the uncertainties of water availability", **it does not attribute water scarcity specifically to declining groundwater levels.** Therefore, assumption made in statement I is incorrect.
Statement II is incorrect. Though the passage states that large irrigation projects often dominate public discourse and that nearly 52 per cent of the country's gross cropped area remains rainfed, **it does not discuss the status of their (**large irrigation projects**) implementation or their efficiency.** Therefore, assumption made in statement II is incorrect.
Subject:) CSAT
Topic:) Reading Comprehension
Subtopic:)